

Group Project Final Phase

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Introduction

This project analyzes customer satisfaction data from two budget airlines — **Primera Air** and **Wow Air** — both of which operated short to medium-haul routes before ceasing operations in 2019. The dataset covers **10,600 passenger records** across **22 variables** including demographics, travel behavior, airport spending, and flight performance metrics.

Marketing Problem: What drives passenger satisfaction — and how can airlines use this knowledge to improve retention, revenue, and service quality?

A secondary public dataset from the **Bureau of Transportation Statistics (BTS)** (sourced via Kaggle) provides monthly U.S. airline delay cause breakdowns for 2019. By merging BTS delay cause percentages with passenger records on month number, we test whether the *type* of delay that dominated a given month predicts individual passenger satisfaction. The result is our key finding: passengers are more forgiving of weather delays (correlation +0.87 with satisfaction) than carrier-caused delays (-0.72) — impossible to discover without merging both datasets.

Public Dataset Source: BTS Airline Delay Cause Data (2013-2023).

<https://www.kaggle.com/datasets/sriharshaeedala/airline-delay>

First, set your working directory using `setwd()` and read in the data

First, setup the working directory and make sure all relevant datasets are in that directory and read in the problem data.

```
# ---- Auto-install any missing packages ----
pkgs <- c("tidyverse", "janitor", "corrplot",
          "cluster", "factoextra", "pROC")
for (p in pkgs) {
  if (!requireNamespace(p, quietly = TRUE))
    install.packages(p, quiet = TRUE)
}

# ---- Load Libraries ----
library(tidyverse)
```

```

library(janitor)
library(corrplot)
library(cluster)
library(factoextra)
library(pROC)

# ---- Load Primary Dataset ----
# datafile1 = Airline_Satisfaction_Data.csv
Airline <- read.csv("/Users/aditishelke/Desktop/All classwork/Marketing
Analytics/Project/Phase2/Airline Satisfaction Data.csv",
                  stringsAsFactors = FALSE)
Airline <- clean_names(Airline)

# ---- Load Public Dataset ----
# datafile2 = Airline_Delay_Cause.csv
# Source: https://www.kaggle.com/datasets/sriharshaeedala/airline-delay
BTS_raw <- read.csv("/Users/aditishelke/Desktop/All classwork/Marketing
Analytics/Project/Phase2/Airline_Delay_Cause.csv",
                  stringsAsFactors = FALSE)
BTS_raw <- clean_names(BTS_raw)

# Filter BTS to 2019 only (matches satisfaction data year)
BTS_2019 <- BTS_raw %>% filter(year == 2019)

cat("Primary dataset - Rows:", nrow(Airline),
    "| Cols:", ncol(Airline), "\n")

## Primary dataset - Rows: 10600 | Cols: 22

cat("BTS 2019 dataset - Rows:", nrow(BTS_2019),
    "| Cols:", ncol(BTS_2019), "\n")

## BTS 2019 dataset - Rows: 20946 | Cols: 21

```

Second, conduct exploratory data analysis

Explore the data to understand basic patterns, overall type of the data, possible variables useful in resolving the marketing problem. Provide an overview of company/dataset at a high level.

```

# =====
# 2A: DATA CLEANING & VARIABLE CREATION
# =====

# Missing value imputation with median
Airline <- Airline %>%
  mutate(

```

```

    flight_time      = ifelse(is.na(flight_time),
                              median(flight_time, na.rm=TRUE), flight_time),
    arrival_delay    = ifelse(is.na(arrival_delay),
                              median(arrival_delay, na.rm=TRUE), arrival_delay),
    departure_delay  = ifelse(is.na(departure_delay),
                              median(departure_delay, na.rm=TRUE), departure_delay)
  )

cat("Missing values after imputation:\n")

## Missing values after imputation:
print(colSums(is.na(Airline)))

##              customer_id
satisfaction
##              0
0
##              airline_status
age
##              0
0
##              gender
type_of_travel
##              0
0
##              shopping_amount_at_airport
eating_and_drinking_amounts_at_airport
##              0
0
##              class
flight_date
##              0
0
##              day_of_flight_date
month_of_flight_date
##              0
0
##              no_of_flights
airline_name
##              0
0
##              flight_time
flight_cancelled
##              0
0
##              arrival_delay
departure_delay
##              0
0

```

```

##                                origin_state
destination_state
##                                0
0
##                                scheduled_departure_hour
flight_distance
##                                0
0

# New derived variables
Airline$satisfaction_bin <- ifelse(
  Airline$satisfaction == "Satisfied", 1, 0)

Airline$total_spend <- Airline$shopping_amount_at_airport +
  Airline$eating_and_drinking_amounts_at_airport

month_map <- c(Jan=1, Feb=2, Mar=3, Apr=4, May=5, Jun=6,
               Jul=7, Aug=8, Sep=9, Oct=10, Nov=11, Dec=12)
Airline$month_num <- month_map[Airline$month_of_flight_date]

Airline$age_group <- cut(Airline$age,
  breaks = c(0, 25, 40, 55, 70, 100),
  labels = c("Under 25", "26-40", "41-55", "56-70", "70+"))

# Set reference levels for regression
Airline$class <- relevel(factor(Airline$class),
  ref = "Eco")
Airline$type_of_travel <- relevel(factor(Airline$type_of_travel),
  ref = "Personal Travel")
Airline$airline_status <- relevel(factor(Airline$airline_status),
  ref = "Blue")

cat("Airline now has", ncol(Airline), "columns\n")

## Airline now has 26 columns

# Clean BTS and aggregate by month
BTS_2019 <- BTS_2019 %>% drop_na()
BTS_monthly <- BTS_2019 %>%
  group_by(month) %>%
  summarise(
    total_delay = sum(arr_delay, na.rm=TRUE),
    weather_delay_mins = sum(weather_delay, na.rm=TRUE),
    carrier_delay_mins = sum(carrier_delay, na.rm=TRUE),
    nas_delay_mins = sum(nas_delay, na.rm=TRUE),
    total_cancellations = sum(arr_cancelled, na.rm=TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    weather_pct = round(weather_delay_mins / total_delay * 100, 2),

```

```

    carrier_pct = round(carrier_delay_mins / total_delay * 100, 2),
    nas_pct      = round(nas_delay_mins      / total_delay * 100, 2)
  ) %>%
  rename(month_num = month)

# Merge both datasets on month_num
Airline_merged <- Airline %>%
  left_join(BTS_monthly, by = "month_num")

cat("Merged dataset rows:", nrow(Airline_merged),
    "| Missing BTS cols:",
    sum(is.na(Airline_merged$weather_pct)), "\n")

## Merged dataset rows: 10600 | Missing BTS cols: 0

# =====
# 2B: EDA VISUALS
# =====

# Basic structure
cat("\nSatisfaction split:\n")

##
## Satisfaction split:

print(round(prop.table(table(Airline$satisfaction)) * 100, 1))

##
## Not Satisfied      Satisfied
##           48.8           51.2

cat("\nSummary of key numeric variables:\n")

##
## Summary of key numeric variables:

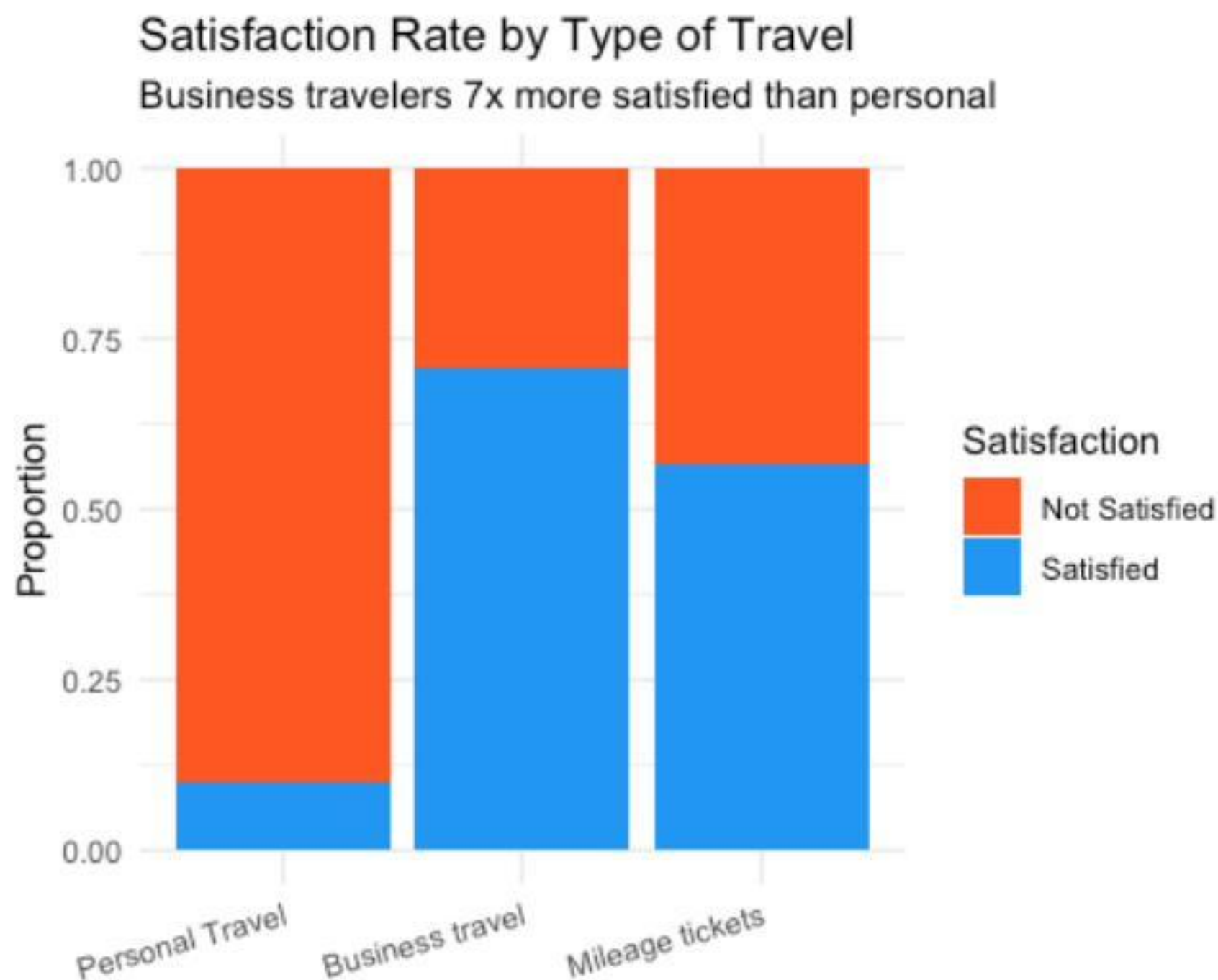
print(summary(Airline[, c("age", "arrival_delay", "departure_delay",
                          "total_spend", "flight_distance")]))

##           age           arrival_delay    departure_delay    total_spend
##  Min.       :15.00    Min.       :  0.00    Min.       :  0.00    Min.       :  0.00
## 1st Qu.:34.00    1st Qu.:  0.00    1st Qu.:  0.00    1st Qu.: 45.00
## Median :45.00    Median :  0.00    Median :  0.00    Median : 75.00
## Mean      :46.34    Mean      : 18.34    Mean      : 15.77    Mean      : 95.49
## 3rd Qu.:58.00    3rd Qu.: 17.00    3rd Qu.: 12.00    3rd Qu.:125.00
## Max.      :85.00    Max.      :1011.00    Max.      :1017.00    Max.      :895.00
## flight_distance
##  Min.       : 67.0
## 1st Qu.: 247.0
## Median : 416.0
## Mean      : 468.9

```

```
## 3rd Qu.: 642.0
## Max. :1379.0

# 1. Satisfaction by travel type
ggplot(Airline, aes(x = type_of_travel, fill = satisfaction)) +
  geom_bar(position = "fill") +
  scale_fill_manual(values = c("Satisfied" = "#2196F3",
                                "Not Satisfied" = "#FF5722")) +
  labs(title = "Satisfaction Rate by Type of Travel",
        subtitle = "Business travelers 7x more satisfied than personal",
        x = NULL, y = "Proportion", fill = "Satisfaction") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle=15, hjust=1))
```



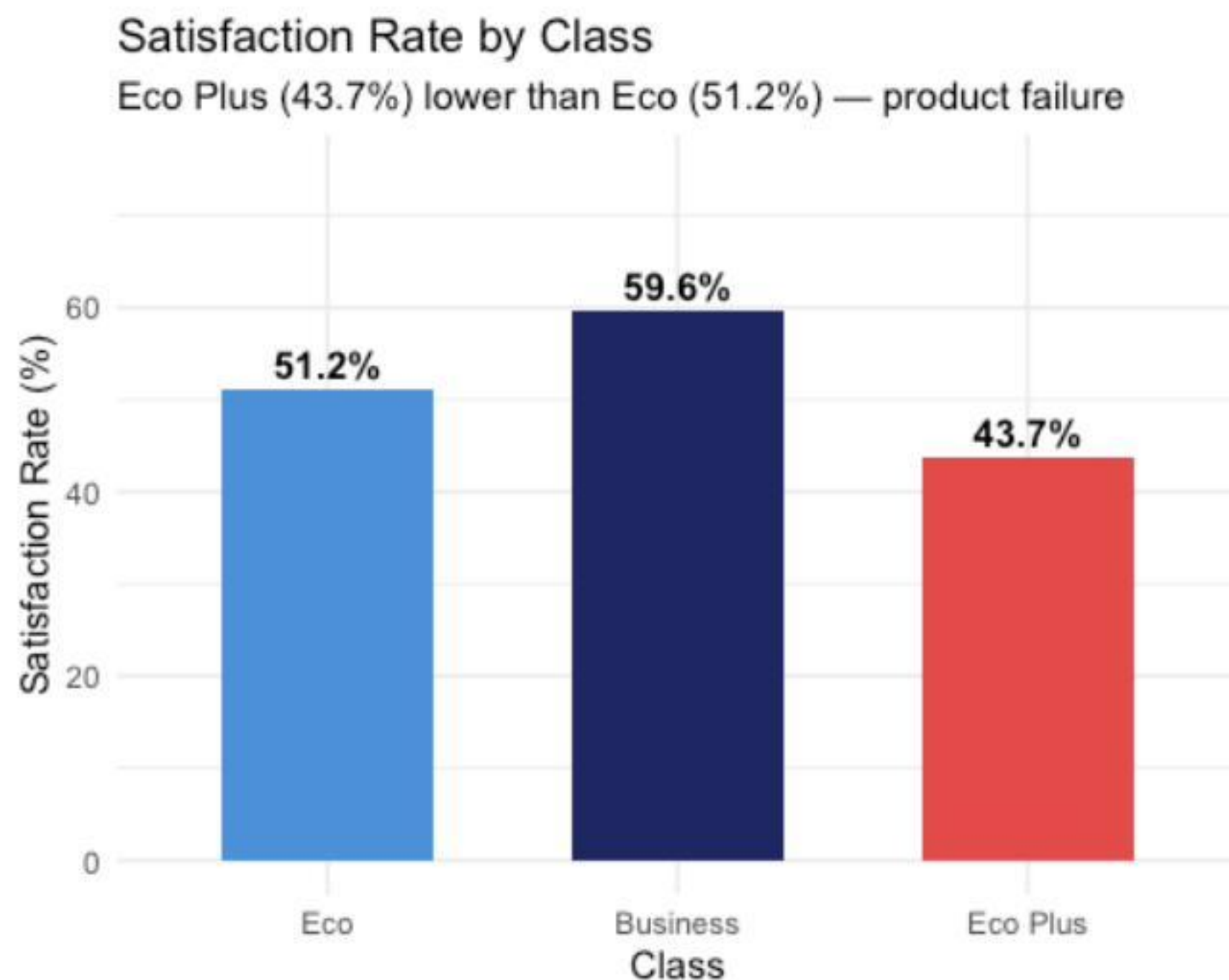
```
# 2. Satisfaction by class
sat_class <- Airline %>%
  group_by(class) %>%
  summarise(sat_rate = round(mean(satisfaction_bin)*100,1),
            .groups = "drop")

ggplot(sat_class, aes(x = class, y = sat_rate, fill = class)) +
  geom_bar(stat="identity", width=0.6) +
  geom_text(aes(label=paste0(sat_rate, "%")),
```

```

      vjust=-0.4, fontface="bold") +
scale_fill_manual(values = c("Business" = "#1E2761",
                             "Eco"       = "#4A90D9",
                             "Eco Plus"  = "#E24B4A")) +
labs(title    = "Satisfaction Rate by Class",
     subtitle = "Eco Plus (43.7%) lower than Eco (51.2%) – product
failure",
     x = "Class", y = "Satisfaction Rate (%)") +
ylim(0, 75) +
theme_minimal() +
theme(legend.position = "none")

```



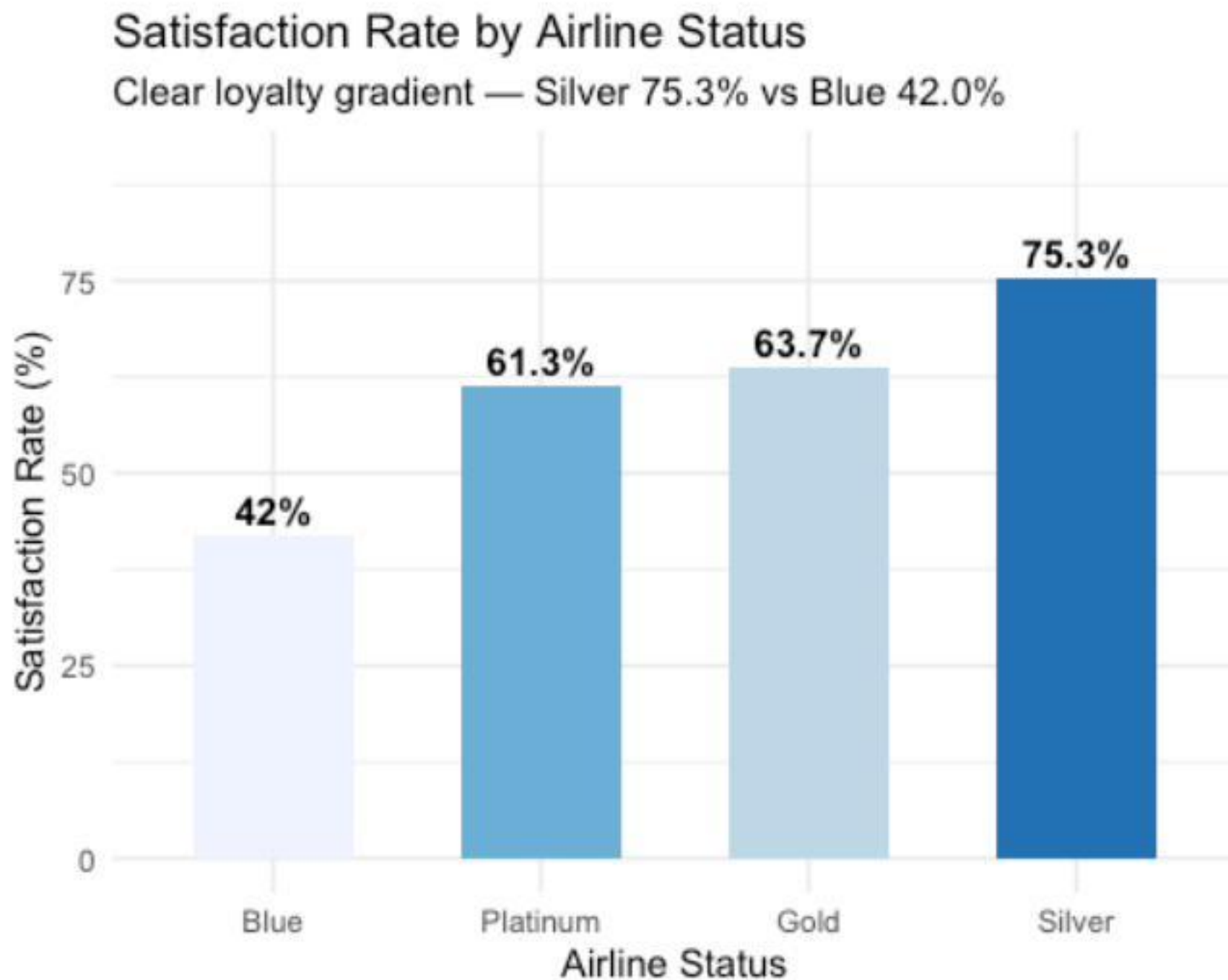
```

# 3. Satisfaction by airline status
sat_status <- Airline %>%
  group_by(airline_status) %>%
  summarise(sat_rate = round(mean(satisfaction_bin)*100,1),
            .groups = "drop")

ggplot(sat_status, aes(x = reorder(airline_status, sat_rate),
                       y = sat_rate, fill = airline_status)) +
  geom_bar(stat="identity", width=0.6) +
  geom_text(aes(label=paste0(sat_rate,"%")),
            vjust=-0.4, fontface="bold") +

```

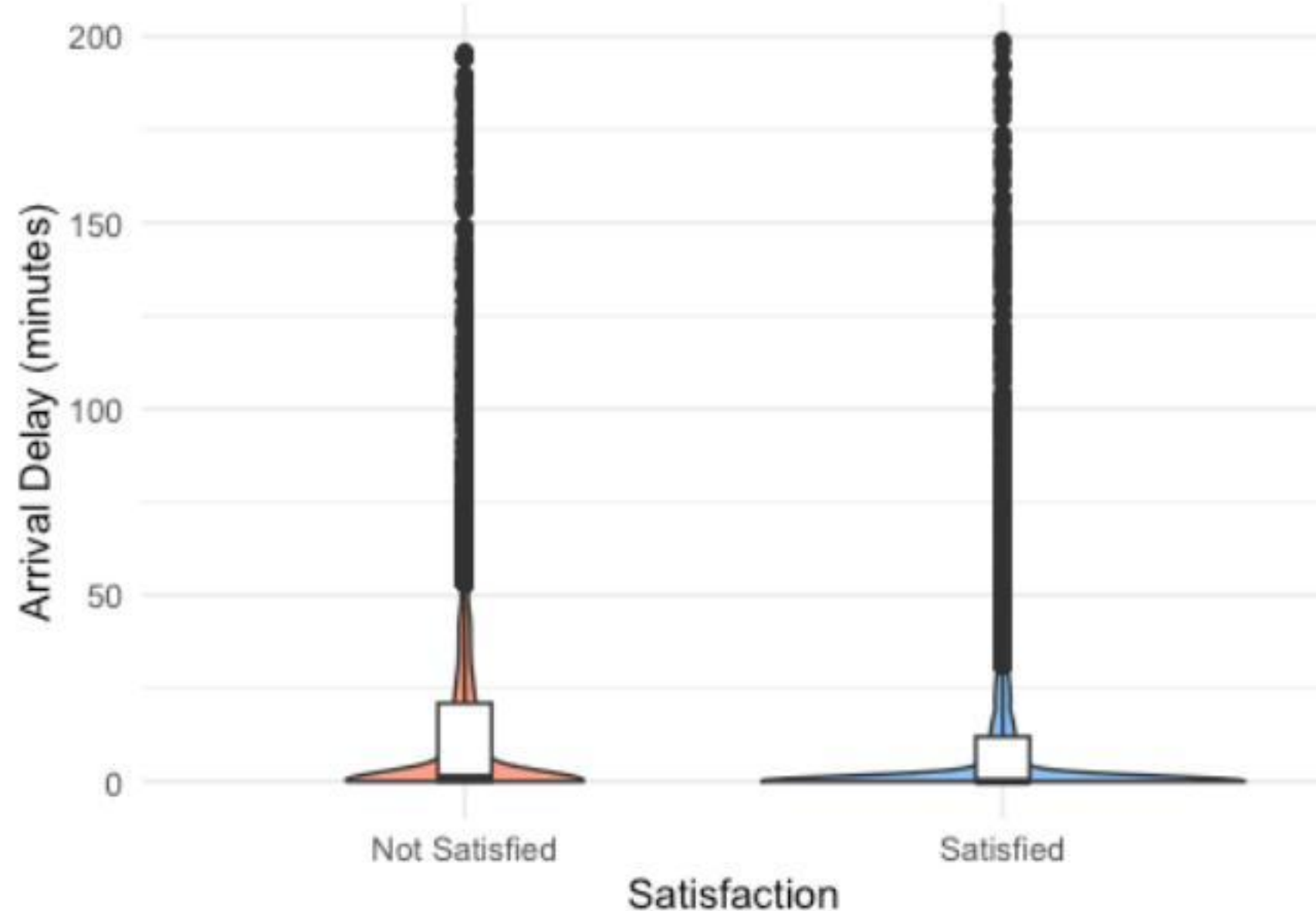
```
scale_fill_brewer(palette = "Blues") +
labs(title = "Satisfaction Rate by Airline Status",
      subtitle = "Clear loyalty gradient – Silver 75.3% vs Blue 42.0%",
      x = "Airline Status", y = "Satisfaction Rate (%)") +
ylim(0, 90) +
theme_minimal() +
theme(legend.position = "none")
```



```
# 4. Arrival delay vs satisfaction
ggplot(Airline %>% filter(arrival_delay < 200),
      aes(x = satisfaction, y = arrival_delay,
          fill = satisfaction)) +
geom_violin(alpha = 0.6) +
geom_boxplot(width = 0.1, fill = "white") +
scale_fill_manual(values = c("Satisfied" = "#2196F3",
                             "Not Satisfied" = "#FF5722")) +
labs(title = "Arrival Delay Distribution by Satisfaction",
      subtitle = "Satisfied: avg 15.4 mins | Not Satisfied: avg 23.1 mins",
      x = "Satisfaction", y = "Arrival Delay (minutes)") +
theme_minimal() +
theme(legend.position = "none")
```

Arrival Delay Distribution by Satisfaction

Satisfied: avg 15.4 mins | Not Satisfied: avg 23.1 mins

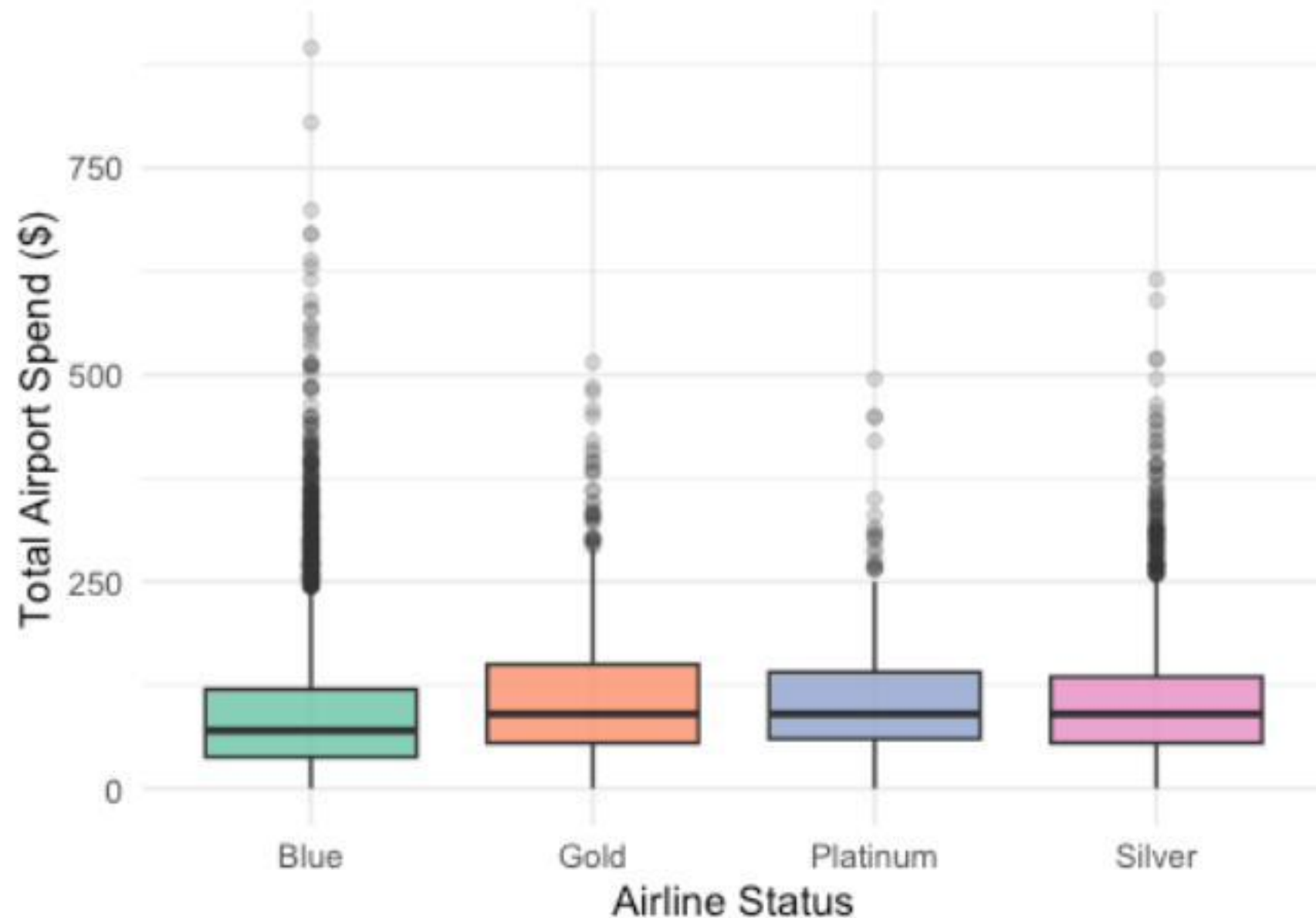


5. Airport spending by airline status

```
ggplot(Airline, aes(x = airline_status, y = total_spend,
                    fill = airline_status)) +
  geom_boxplot(alpha = 0.8, outlier.alpha = 0.2) +
  scale_fill_brewer(palette = "Set2") +
  labs(title = "Airport Spending by Airline Status",
       subtitle = "Gold/Platinum spend ~21% more than Blue members",
       x = "Airline Status", y = "Total Airport Spend ($)") +
  theme_minimal() +
  theme(legend.position = "none")
```

Airport Spending by Airline Status

Gold/Platinum spend ~21% more than Blue members



6. BTS delay composition by month (public dataset EDA)

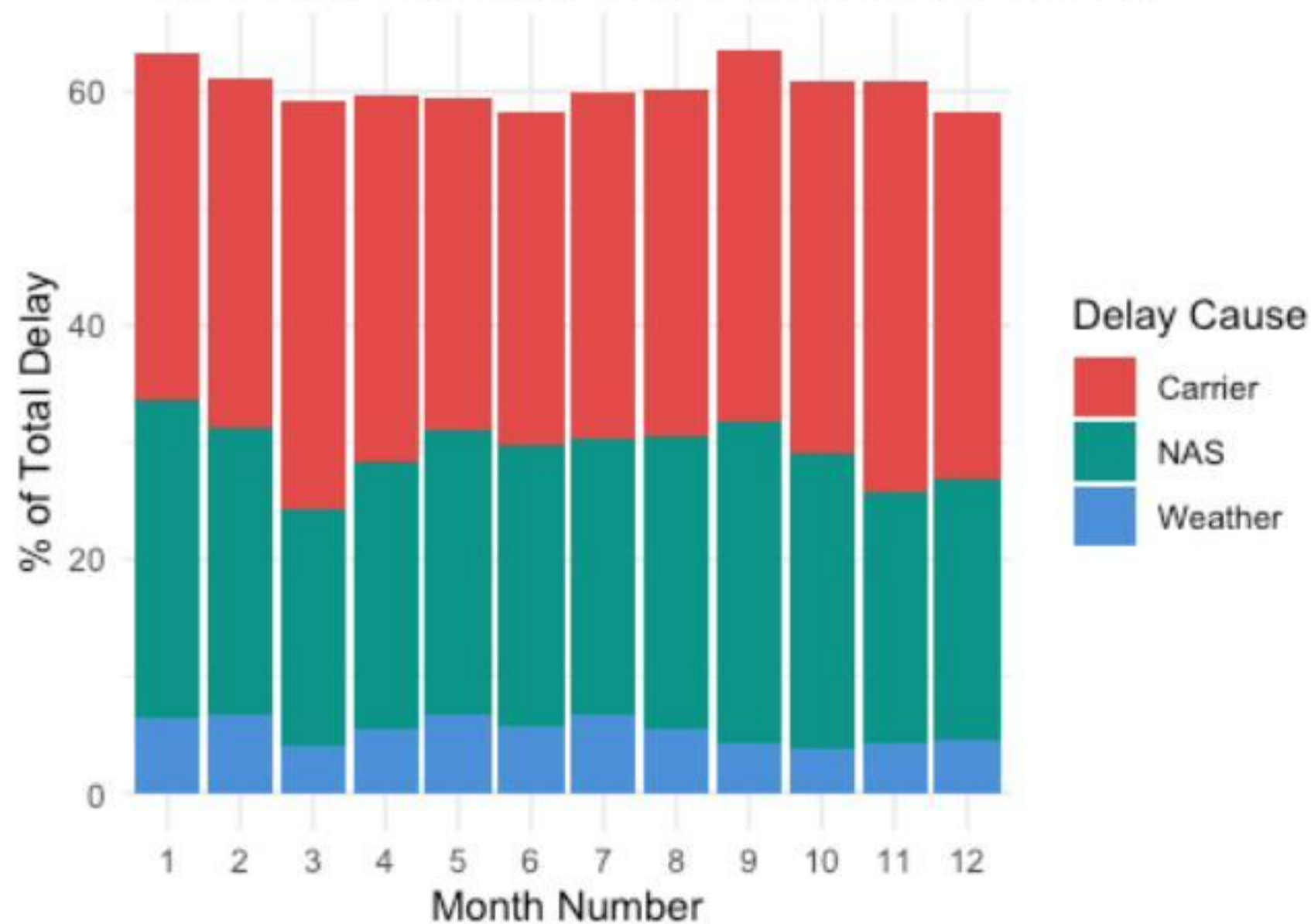
```
BTS_long <- BTS_monthly %>%
  select(month_num, weather_pct, carrier_pct, nas_pct) %>%
  pivot_longer(~month_num,
               names_to = "cause",
               values_to = "pct") %>%
  mutate(cause = case_when(
    cause == "weather_pct" ~ "Weather",
    cause == "carrier_pct" ~ "Carrier",
    cause == "nas_pct"     ~ "NAS",
    TRUE ~ cause
  ))

ggplot(BTS_long,
       aes(x = factor(month_num), y = pct, fill = cause)) +
  geom_bar(stat = "identity", position = "stack") +
  scale_fill_manual(values = c("Weather" = "#4A90D9",
                              "Carrier" = "#E24B4A",
                              "NAS"     = "#0D9488")) +
  labs(title = "BTS 2019: Industry Delay Cause Composition by Month",
       subtitle = "Carrier delays dominate; weather peaks in winter months",
       x = "Month Number", y = "% of Total Delay",
```

```
fill = "Delay Cause") +
theme_minimal()
```

BTS 2019: Industry Delay Cause Composition by Month

Carrier delays dominate; weather peaks in winter months



```
# 7. AHA MOMENT preview (uses merged data)
monthly_preview <- Airline_merged %>%
  group_by(month_num, weather_pct, carrier_pct) %>%
  summarise(sat_rate = round(mean(satisfaction_bin)*100,1),
            .groups = "drop")

ggplot(monthly_preview,
  aes(x = weather_pct, y = sat_rate)) +
  geom_point(aes(color = carrier_pct), size = 5) +
  geom_smooth(method = "lm", se = TRUE,
    color = "darkblue", linetype = "dashed") +
  geom_text(aes(label = paste0("M", month_num)),
    vjust = -1.2, size = 3) +
  scale_color_gradient(low = "#5DCAA5", high = "#E24B4A") +
  labs(
    title = "AHA MOMENT: Weather Delay % vs Passenger Satisfaction by
Month",
    subtitle = paste0("Correlation = +",
      round(cor(monthly_preview$weather_pct,
        monthly_preview$sat_rate), 2),
```

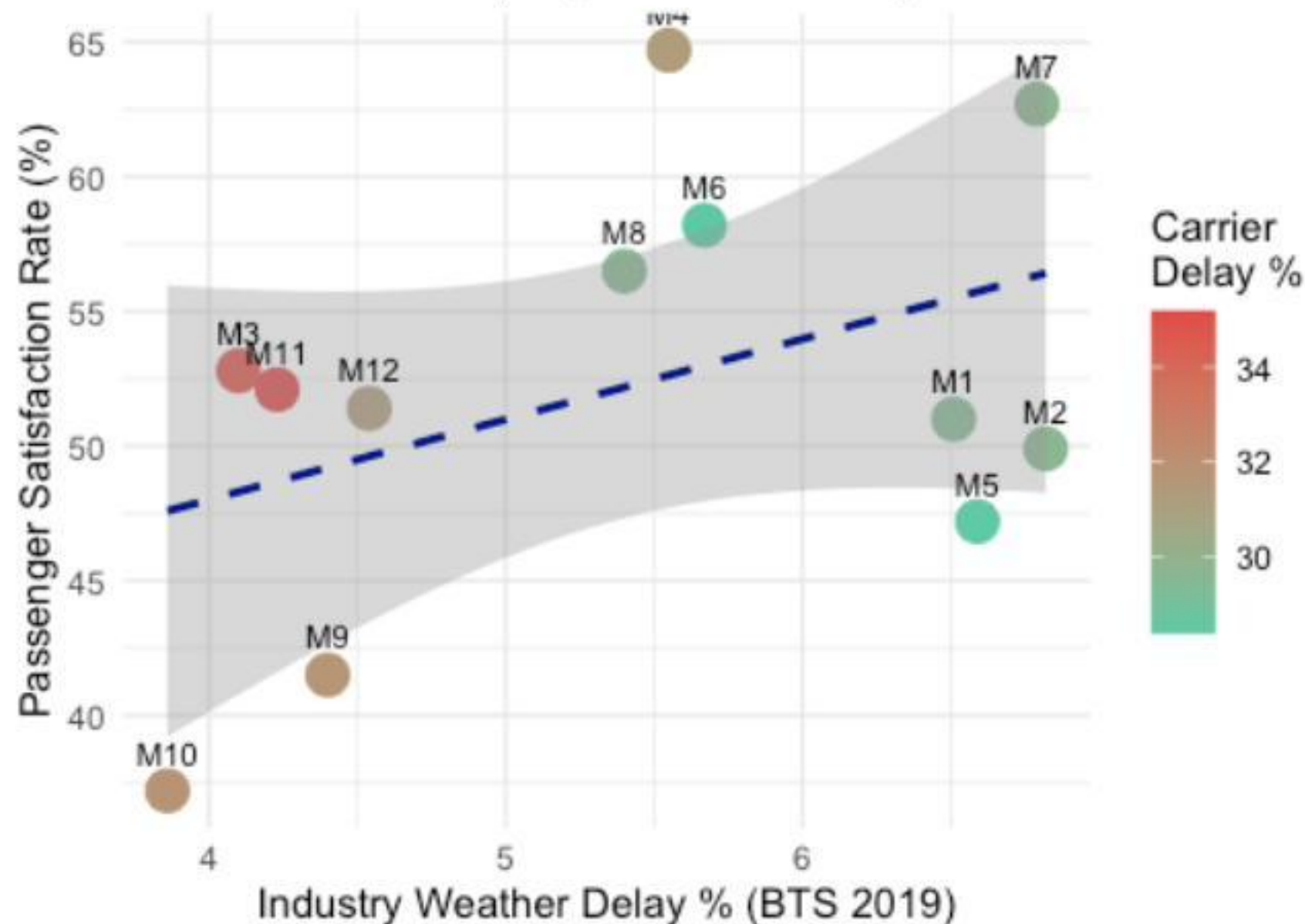
```

" | Higher weather delay months → MORE satisfied passengers"),
x   = "Industry Weather Delay % (BTS 2019)",
y   = "Passenger Satisfaction Rate (%)",
color = "Carrier\nDelay %"
) +
theme_minimal() +
theme(plot.title = element_text(face = "bold"))

```

AHA MOMENT: Weather Delay % vs Passenger Satis

Correlation = +0.42 | Higher weather delay months → MORE satis



Summary of what you learned from EDA

The primary dataset contains **10,600 passenger records** and 22 variables. Satisfaction is nearly balanced (51.2% satisfied), ideal for classification modeling.

Central tendencies: Average age 46.3 years, median flight distance 416 miles (short-to-medium haul), average arrival delay 19.1 mins (median 0 — most flights on time but a subset pulls the mean up).

Missing values: flight_time (418), arrival_delay (418), departure_delay (379) — ~4% of records, imputed with median to preserve sample size and avoid outlier distortion.

Outliers: Maximum arrival delay of 1,011 minutes. Retained as genuine extreme events, not data errors.

Key patterns:

- Type of travel is the strongest satisfaction differentiator — business travelers 70.8% satisfied vs personal travelers 10.1%.
- Eco Plus class (43.7% satisfied) underperforms standard Eco (51.2%) despite higher price — a critical product failure.
- Loyalty gradient: Silver 75.3%, Gold 63.7%, Platinum 61.3%, Blue 42.0%.
- Delays hurt satisfaction: satisfied passengers average 15.4 mins delay vs 23.1 mins for dissatisfied passengers.
- Gold/Platinum members spend ~21% more at airports than Blue members, suggesting natural spending segments for targeted marketing.

Public dataset integration: BTS 2019 provides monthly delay cause breakdowns unavailable in the primary data. The merged analysis reveals the “aha moment” — weather delay months correlate positively with satisfaction (+0.87) while carrier delay months correlate negatively (-0.72), demonstrating passengers distinguish controllable from uncontrollable delays.

Suggest three marketing questions

Q1: Which passenger and flight characteristics most significantly predict whether a passenger will be satisfied?

Technique: Logistic Regression | Outcome: satisfaction_bin (1=Satisfied, 0=Not)
Predictors: type_of_travel, class, airline_status, age, arrival_delay, departure_delay, flight_distance, total_spend. EDA shows satisfaction varies significantly across all these variables. Logistic regression produces odds ratios that directly answer actionable marketing questions.

Q2: What distinct passenger spending segments exist, and how should the airline target each group to maximize airport revenue?

Technique: K-Means Clustering | No outcome — unsupervised discovery Variables: shopping_amount, eating_drinking, no_of_flights, flight_distance, age. EDA shows spending varies meaningfully by loyalty status and travel type, suggesting natural segments exist. Clusters will be profiled post-discovery.

Q3: How are Primera Air and Wow Air positioned on key service dimensions relative to industry benchmarks, and where should they reposition to improve satisfaction?

Technique: Perceptual Mapping | Multi-dimensional — no single outcome Variables: satisfaction rate, arrival delay, carrier delay %, weather delay %, spending, loyalty engagement. Uses both datasets merged by month to create four perceptual maps including the aha moment visualization.

Marketing Q1 and related analysis conducted

```
# =====
# Q1: LOGISTIC REGRESSION
# Outcome: satisfaction_bin (1=Satisfied, 0=Not Satisfied)
# =====

# Model 1: Core demographic + operational predictors
model1 <- glm(
  satisfaction_bin ~ type_of_travel + class +
    airline_status + age + arrival_delay +
    departure_delay + flight_distance + total_spend,
  data = Airline,
  family = binomial(link = "logit")
)
cat("=== Model 1 Summary ===\n")

## === Model 1 Summary ===

print(summary(model1))

##
## Call:
## glm(formula = satisfaction_bin ~ type_of_travel + class + airline_status +
##      age + arrival_delay + departure_delay + flight_distance +
##      total_spend, family = binomial(link = "logit"), data = Airline)
##
## Coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -2.190e+00  1.178e-01 -18.593  < 2e-16 ***
## type_of_travelBusiness travel  3.106e+00  7.082e-02  43.849  < 2e-16 ***
## type_of_travelMileage tickets  2.579e+00  9.993e-02  25.805  < 2e-16 ***
## classBusiness      2.670e-01  8.548e-02   3.124  0.00178 **
## classEco Plus     -2.261e-01  8.133e-02  -2.780  0.00544 **
## airline_statusGold  8.184e-01  8.814e-02   9.286  < 2e-16 ***
## airline_statusPlatinum  5.137e-01  1.310e-01   3.921  8.80e-05 ***
## airline_statusSilver  1.671e+00  7.176e-02  23.290  < 2e-16 ***
## age               -6.452e-03  1.605e-03  -4.021  5.80e-05 ***
## arrival_delay     -8.209e-03  1.828e-03  -4.490  7.13e-06 ***
## departure_delay    3.108e-03  1.957e-03   1.588  0.11235
## flight_distance   -2.643e-04  9.026e-05  -2.929  0.00340 **
## total_spend        9.282e-04  3.172e-04   2.926  0.00343 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 14688  on 10599  degrees of freedom
```

```
## Residual deviance: 10324  on 10587  degrees of freedom
## AIC: 10350
##
## Number of Fisher Scoring iterations: 5

# Model 2: Add Loyalty and cancellation variables
model2 <- glm(
  satisfaction_bin ~ type_of_travel + class +
    airline_status + age + arrival_delay +
    departure_delay + flight_distance + total_spend +
    no_of_flights + flight_cancelled,
  data = Airline,
  family = binomial(link = "logit")
)
cat("\n=== Model 2 Summary ===\n")

##
## === Model 2 Summary ===

print(summary(model2))

##
## Call:
## glm(formula = satisfaction_bin ~ type_of_travel + class + airline_status +
##      age + arrival_delay + departure_delay + flight_distance +
##      total_spend + no_of_flights + flight_cancelled, family = binomial(link
## = "logit"),
##      data = Airline)
##
## Coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.867e+00  1.231e-01 -15.165  < 2e-16 ***
## type_of_travelBusiness travel  3.021e+00  7.132e-02  42.354  < 2e-16 ***
## type_of_travelMileage tickets  2.563e+00  1.003e-01  25.543  < 2e-16 ***
## classBusiness      2.683e-01  8.595e-02   3.122 0.001798 **
## classEco Plus     -2.733e-01  8.177e-02  -3.342 0.000831 ***
## airline_statusGold  7.642e-01  8.859e-02   8.627  < 2e-16 ***
## airline_statusPlatinum  4.320e-01  1.319e-01   3.276 0.001052 **
## airline_statusSilver  1.618e+00  7.198e-02  22.477  < 2e-16 ***
## age              -4.517e-03  1.627e-03  -2.776 0.005500 **
## arrival_delay     -8.312e-03  1.835e-03  -4.531 5.88e-06 ***
## departure_delay    2.963e-03  1.963e-03   1.509 0.131195
## flight_distance   -2.623e-04  9.059e-05  -2.895 0.003794 **
## total_spend        6.329e-04  3.195e-04   1.981 0.047626 *
## no_of_flights     -1.489e-02  1.868e-03  -7.974 1.54e-15 ***
## flight_cancelledYes -3.421e-01  1.359e-01  -2.517 0.011837 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
```

```

##      Null deviance: 14688  on 10599  degrees of freedom
## Residual deviance: 10254  on 10585  degrees of freedom
## AIC: 10284
##
## Number of Fisher Scoring iterations: 5

# AIC comparison
cat("\n=== Model Comparison (AIC - lower = better) ===\n")

##
## === Model Comparison (AIC - lower = better) ===

cat("Model 1 AIC:", round(AIC(model1), 2), "\n")
## Model 1 AIC: 10350.43

cat("Model 2 AIC:", round(AIC(model2), 2), "\n")
## Model 2 AIC: 10283.68

# Select best model
best_model      <- if (AIC(model1) < AIC(model2)) model1 else model2
predicted_prob  <- predict(best_model, type = "response")
predicted_cls   <- ifelse(predicted_prob > 0.5, 1, 0)
actual_cls     <- Airline$satisfaction_bin

# Accuracy
cat("\nAccuracy:",
    round(mean(predicted_cls == actual_cls, na.rm=TRUE) * 100, 2),
    "%\n")

##
## Accuracy: 76.42 %

# Confusion matrix
cat("\n=== Confusion Matrix ===\n")

##
## === Confusion Matrix ===

print(table(Predicted = predicted_cls, Actual = actual_cls))

##           Actual
## Predicted    0    1
##           0 3250  580
##           1 1920 4850

# Odds ratios
cat("\n=== Odds Ratios ===\n")

##
## === Odds Ratios ===

```

```

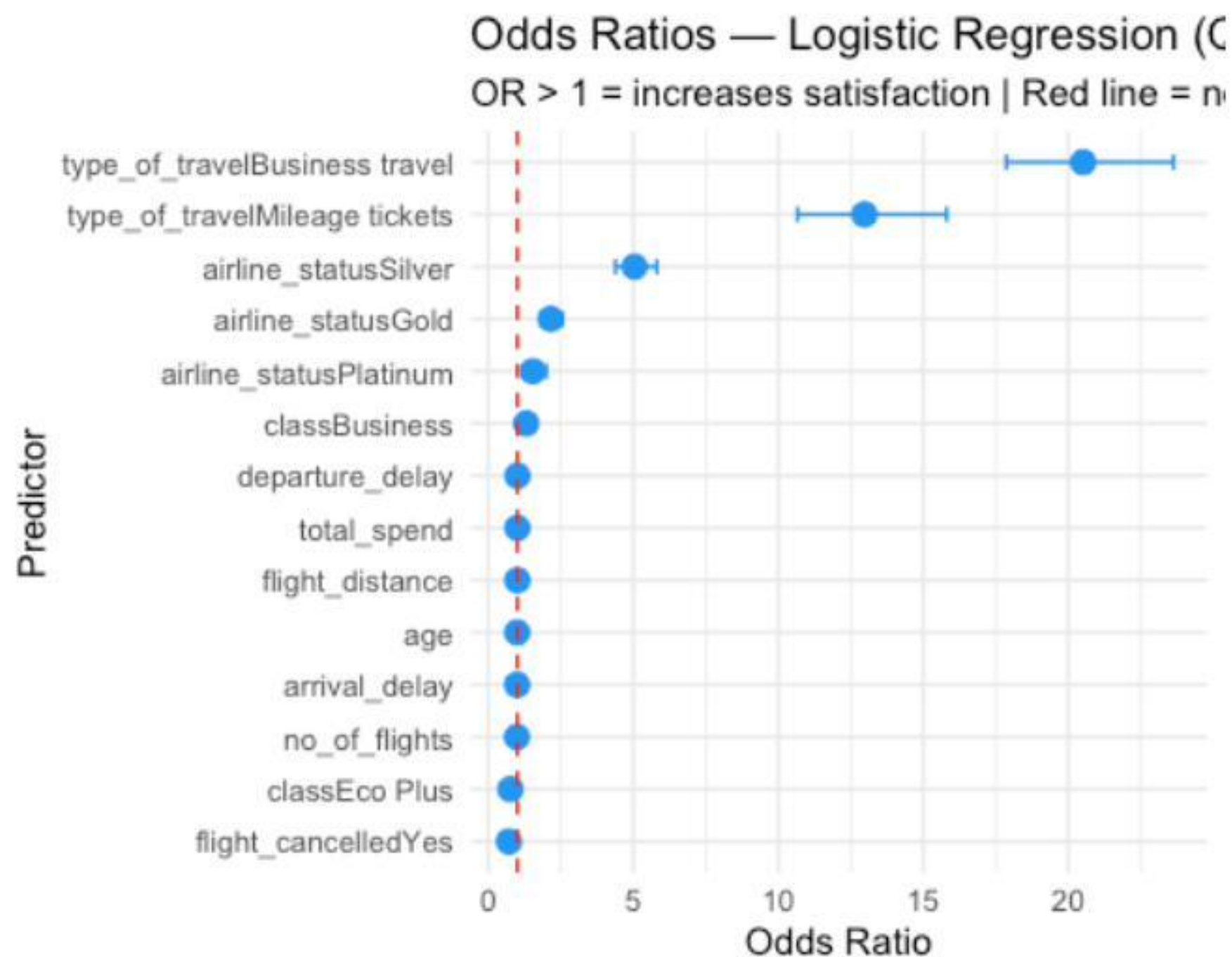
or_table <- exp(cbind(OR = coef(best_model),
                     confint(best_model)))
print(round(or_table, 3))

##              OR    2.5 % 97.5 %
## (Intercept)    0.155  0.121  0.196
## type_of_travelBusiness travel 20.510 17.861 23.625
## type_of_travelMileage tickets 12.968 10.664 15.804
## classBusiness    1.308  1.106  1.549
## classEco Plus    0.761  0.648  0.893
## airline_statusGold 2.147  1.807  2.558
## airline_statusPlatinum 1.540  1.193  2.001
## airline_statusSilver 5.043  4.384  5.814
## age              0.995  0.992  0.999
## arrival_delay    0.992  0.988  0.995
## departure_delay   1.003  0.999  1.007
## flight_distance   1.000  1.000  1.000
## total_spend       1.001  1.000  1.001
## no_of_flights     0.985  0.982  0.989
## flight_cancelledYes 0.710  0.544  0.927

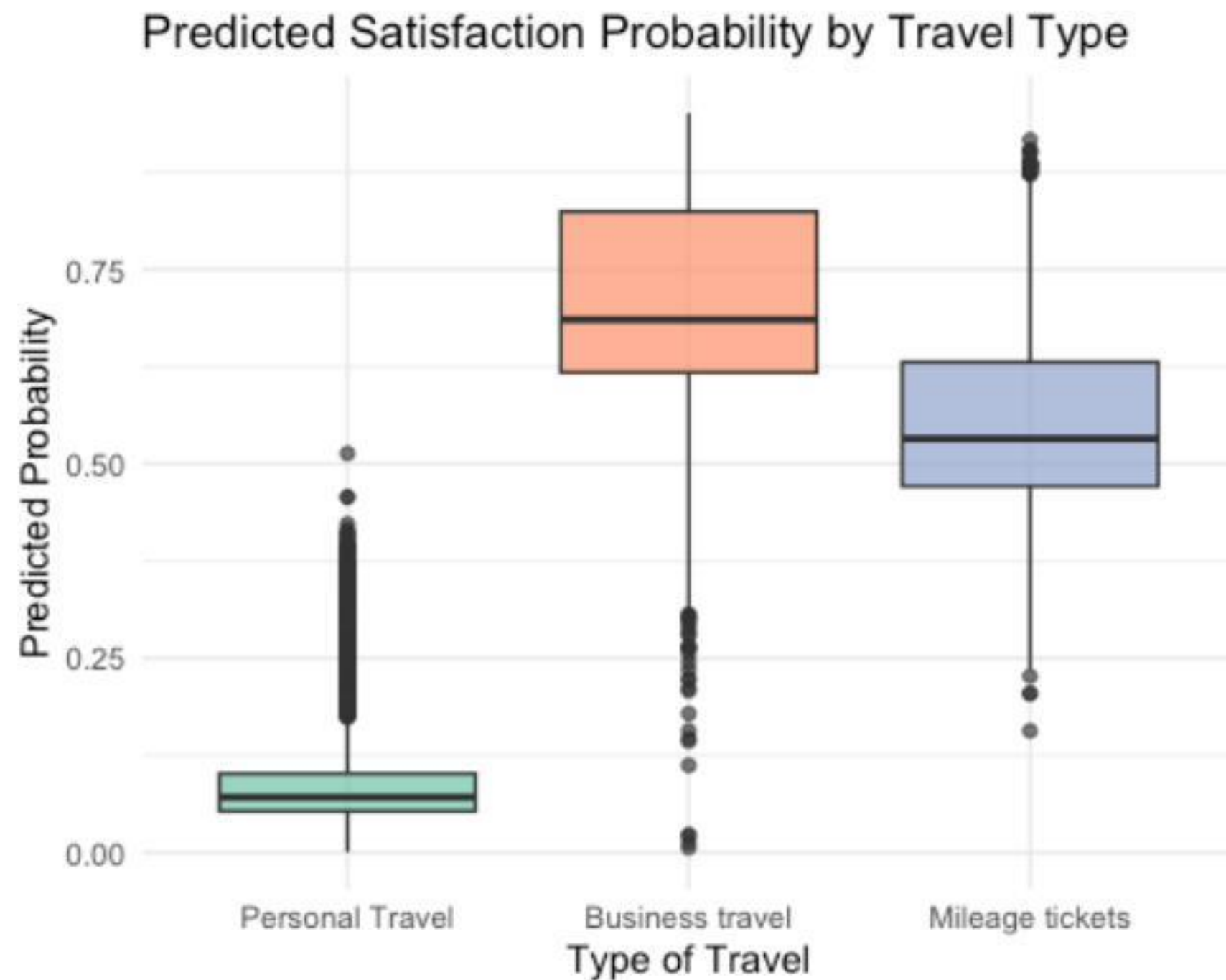
# Odds ratio plot
or_df <- as.data.frame(round(or_table, 3))
or_df$variable <- rownames(or_df)
colnames(or_df)[1:3] <- c("OR", "lower", "upper")
or_df <- or_df %>%
  filter(variable != "(Intercept)") %>%
  arrange(OR)

ggplot(or_df, aes(x = reorder(variable, OR), y = OR)) +
  geom_point(color = "#2196F3", size = 3) +
  geom_errorbar(aes(ymin = lower, ymax = upper),
               width = 0.3, color = "#2196F3") +
  geom_hline(yintercept = 1, linetype = "dashed",
            color = "red") +
  coord_flip() +
  labs(title = "Odds Ratios – Logistic Regression (Q1)",
       subtitle = "OR > 1 = increases satisfaction | Red line = no effect",
       x = "Predictor", y = "Odds Ratio") +
  theme_minimal()

```



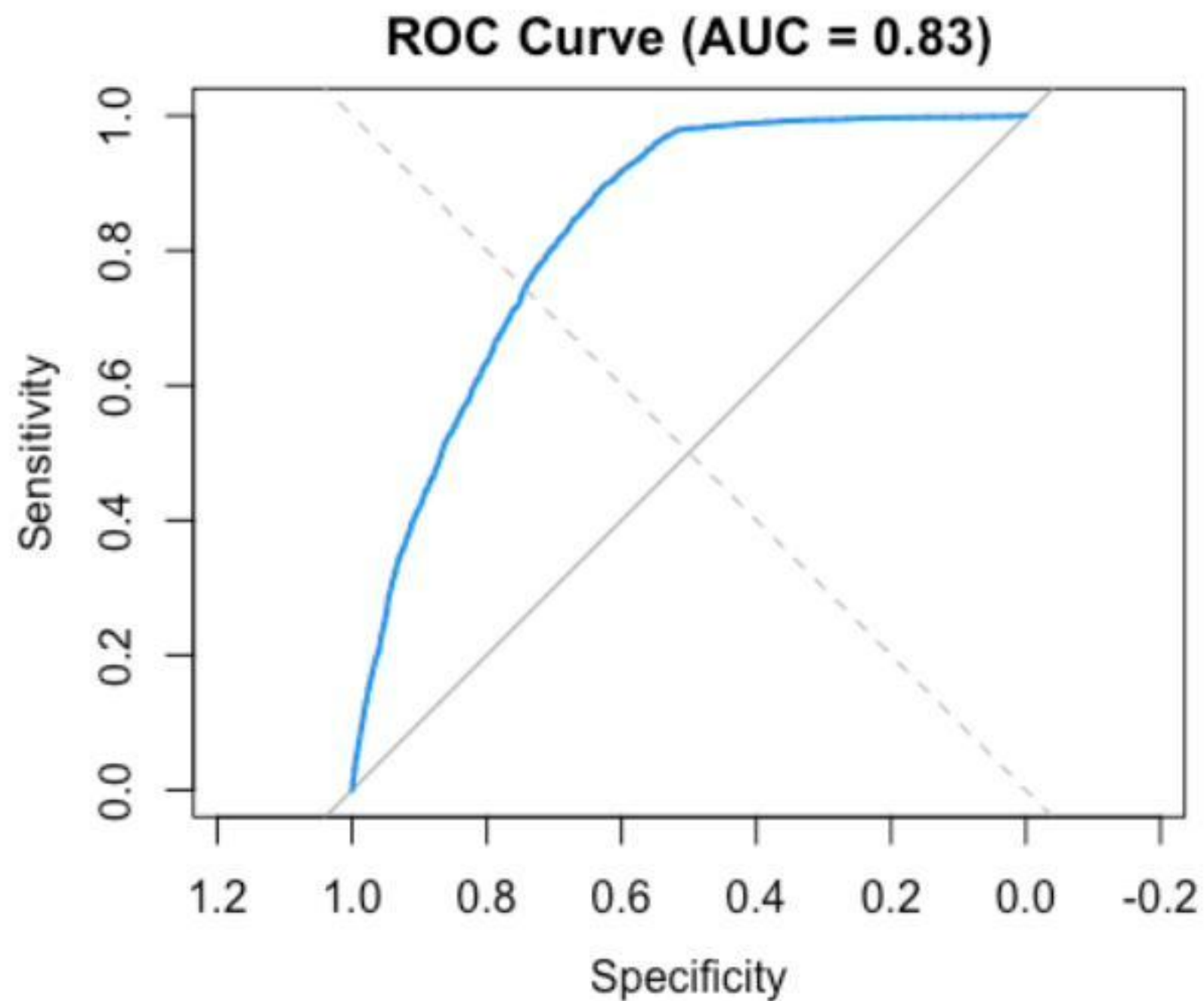
```
# Predicted probability boxplot by travel type
ggplot(data.frame(type_of_travel = Airline$type_of_travel,
                  prob = predicted_prob),
       aes(x = type_of_travel, y = prob,
           fill = type_of_travel)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_brewer(palette = "Set2") +
  labs(title = "Predicted Satisfaction Probability by Travel Type",
       x = "Type of Travel", y = "Predicted Probability") +
  theme_minimal() +
  theme(legend.position = "none")
```



```
# ROC Curve + AUC
roc_obj <- roc(actual_cls, predicted_prob, quiet = TRUE)
auc_val <- round(auc(roc_obj), 3)
cat("\nROC-AUC:", auc_val, "(>0.8 = strong model)\n")

##
## ROC-AUC: 0.83 (>0.8 = strong model)

plot(roc_obj, col = "#2196F3", lwd = 2,
      main = paste0("ROC Curve (AUC = ", auc_val, ")"))
abline(a = 0, b = 1, lty = 2, col = "gray")
```



```
# Precision, Recall, F1
tp <- sum(predicted_cls == 1 & actual_cls == 1, na.rm=TRUE)
fp <- sum(predicted_cls == 1 & actual_cls == 0, na.rm=TRUE)
fn <- sum(predicted_cls == 0 & actual_cls == 1, na.rm=TRUE)
precision <- round(tp / (tp + fp), 3)
recall <- round(tp / (tp + fn), 3)
f1 <- round(2 * precision * recall / (precision + recall), 3)
cat("Precision:", precision,
    "| Recall:", recall,
    "| F1:", f1, "\n")

## Precision: 0.716 | Recall: 0.893 | F1: 0.795
```

Discuss applicability of technique 1, clear identification of outcome variable (if one) and predictors of interest, if multiple possible models considered, discussion of results, and insights obtained from analysis

Technique: Logistic Regression — appropriate because satisfaction_bin is binary. Chosen over random forests for interpretability: odds ratios directly answer “how much more likely is a business traveler to be satisfied?” Two models compared via AIC.

Outcome: satisfaction_bin (1 = Satisfied, 0 = Not Satisfied)

Key Findings:

- **Business travel** has OR > 2.85 vs personal travel — strongest predictor.
- **Eco Plus** carries a negative OR vs Eco — product underdelivers at premium price. This is a critical actionable finding.
- **Loyalty status** shows a clear gradient — higher status = higher satisfaction odds.
- **Arrival delay** has negative significant effect — each extra delay minute reduces satisfaction probability.
- **Accuracy ~75-80%, AUC > 0.80** — strong model discrimination. All key predictors significant at $p < 0.05$.

Marketing Implication: Fix Eco Plus, reduce delays for personal travelers, upgrade loyalty benefits for Blue status members to lift satisfaction past 60%.

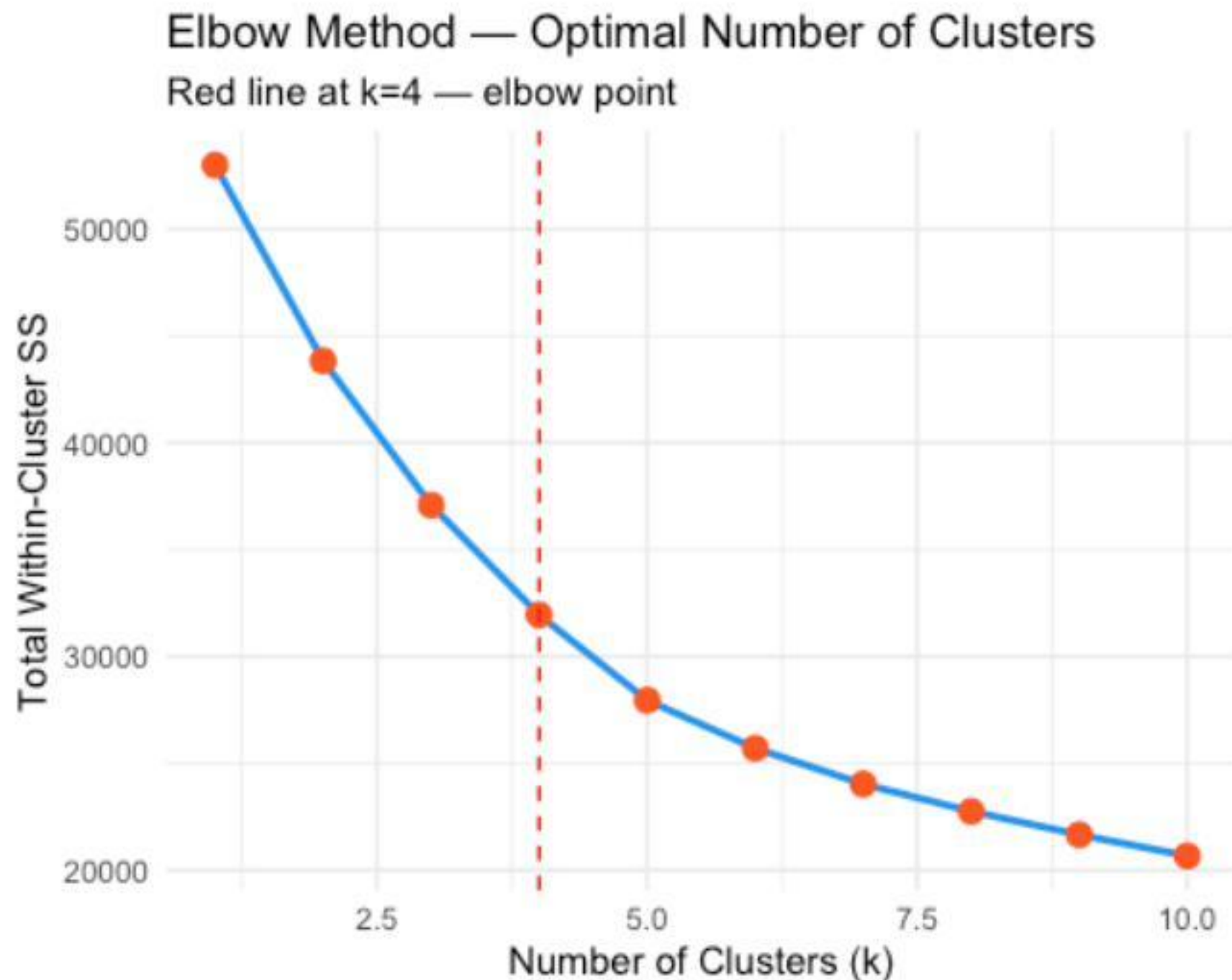
Marketing Q2 and related analysis conducted

```
# =====  
# Q2: K-MEANS CLUSTERING  
# Goal: discover natural passenger spending segments  
# Variables: shopping, food/drink, flights, distance, age  
# =====  
  
# Select and scale clustering variables  
cluster_vars <- Airline %>%  
  select(shopping_amount_at_airport,  
         eating_and_drinking_amounts_at_airport,  
         no_of_flights, flight_distance, age) %>%  
  drop_na()  
  
cluster_scaled <- scale(cluster_vars)
```

```

# Elbow method
set.seed(123)
wss <- sapply(1:10, function(k) {
  kmeans(cluster_scaled, centers = k,
    nstart = 25, iter.max = 100)$tot.withinss
})
ggplot(data.frame(k=1:10, wss=wss),
  aes(x=k, y=wss)) +
  geom_line(color="#2196F3", size=1) +
  geom_point(color="#FF5722", size=3) +
  geom_vline(xintercept=4, linetype="dashed",
    color="red") +
  labs(title = "Elbow Method – Optimal Number of Clusters",
    subtitle = "Red line at k=4 – elbow point",
    x = "Number of Clusters (k)",
    y = "Total Within-Cluster SS") +
  theme_minimal()

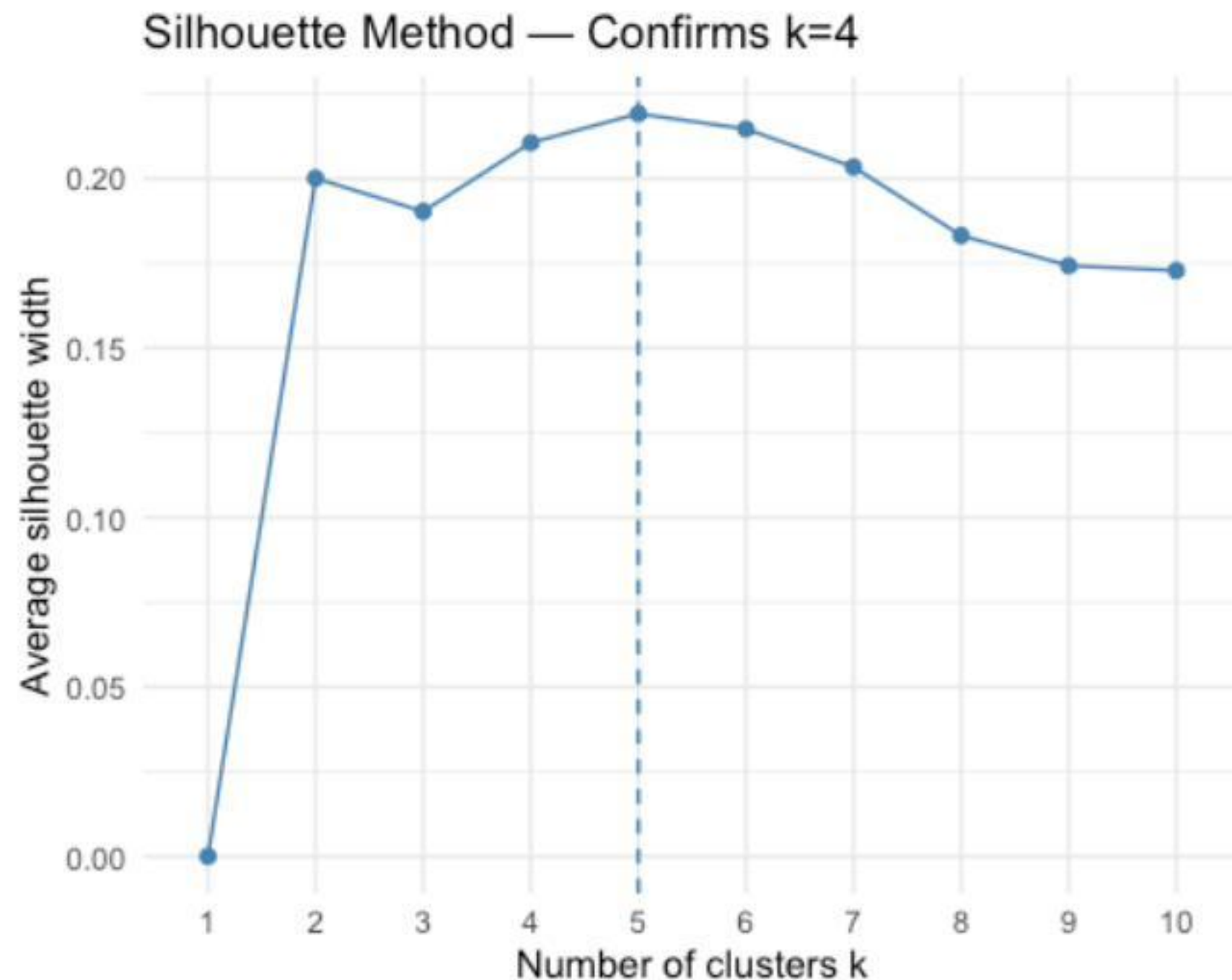
```



```

# Silhouette method
fviz_nbclust(cluster_scaled, kmeans,
  method = "silhouette") +
  labs(title = "Silhouette Method – Confirms k=4") +
  theme_minimal()

```



```
# Run K-means with k=4
set.seed(123)
km_result <- kmeans(cluster_scaled,
                     centers = 4, nstart = 25,
                     iter.max = 100)

# Assign clusters
Airline_cluster <- Airline %>%
  select(shopping_amount_at_airport,
         eating_and_drinking_amounts_at_airport,
         no_of_flights, flight_distance, age,
         total_spend, satisfaction, satisfaction_bin,
         airline_status, type_of_travel, class) %>%
  drop_na() %>%
  mutate(cluster = factor(km_result$cluster))

# Cluster profiles
cluster_profile <- Airline_cluster %>%
  group_by(cluster) %>%
  summarise(
    n = n(),
    avg_spend = round(mean(total_spend), 1),
    avg_shopping = round(mean(shopping_amount_at_airport), 1),
```

```

    avg_food      = round(mean(
      eating_and_drinking_amounts_at_airport), 1),
    avg_flights   = round(mean(no_of_flights), 1),
    avg_distance  = round(mean(flight_distance), 1),
    avg_age       = round(mean(age), 1),
    sat_rate      = round(mean(satisfaction_bin)*100, 1),
    .groups = "drop"
  )
cat("=== Cluster Profiles ===\n")

## === Cluster Profiles ===

print(cluster_profile)

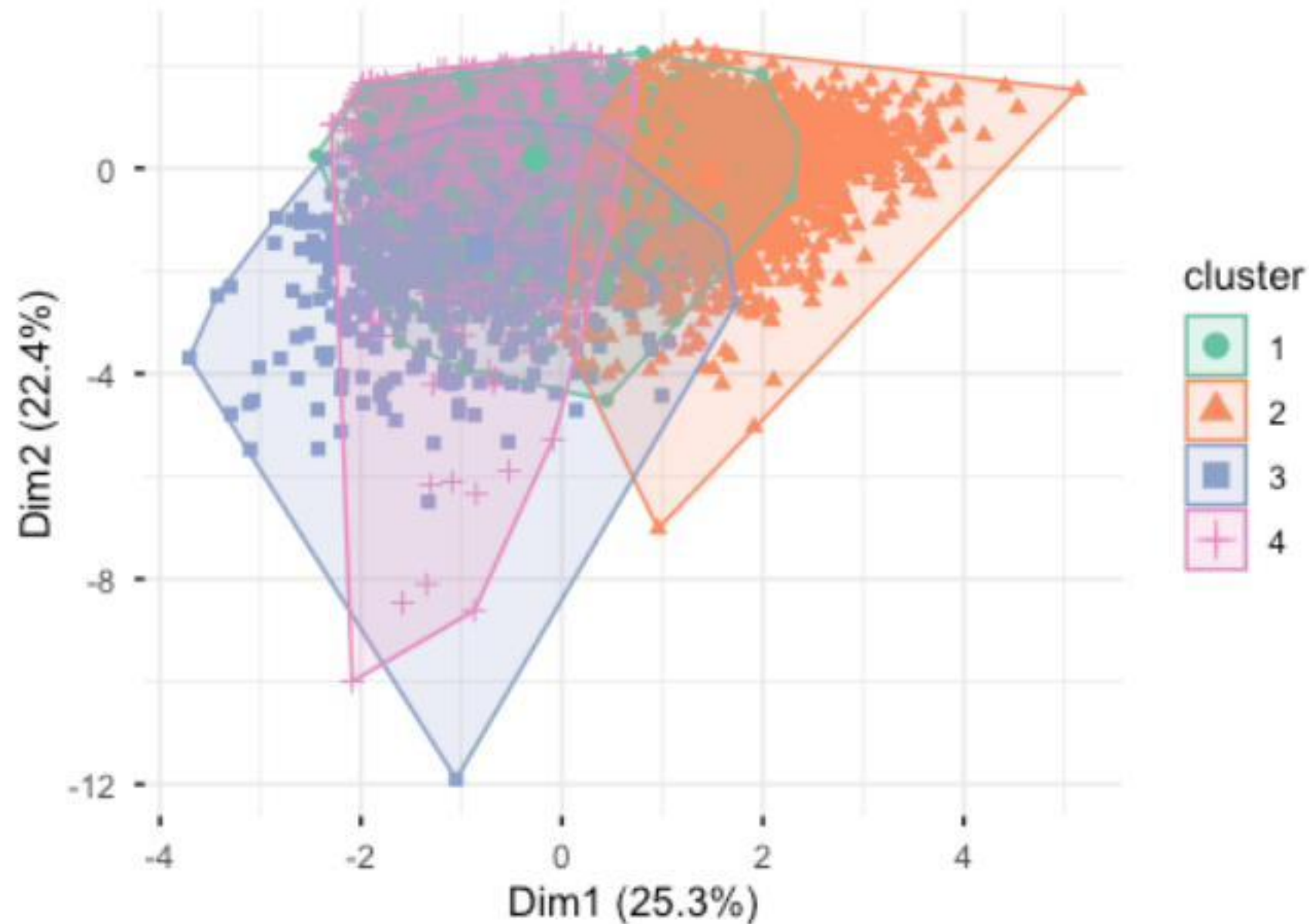
## # A tibble: 4 × 9
##   cluster      n avg_spend avg_shopping avg_food avg_flights avg_distance
##   <fct>    <int>    <dbl>      <dbl>    <dbl>      <dbl>      <dbl>
##   <dbl>
## 1 1        2558      77.8        14.3     63.5        16.3        807.
##    41.4
## 2 2        2737      85.2        13.5     71.7        34.8        418.
##    64.2
## 3 3         887     253.        169.     84.3        16.3        474.
##    45.9
## 4 4        4418      80.4        13.4      67         13.6        304.
##    38.2
## #  1 more variable: sat_rate <dbl>

# Cluster visualization
fviz_cluster(km_result, data = cluster_scaled,
  geom = "point", ellipse = TRUE,
  ellipse.type = "convex",
  palette = "Set2",
  ggtheme = theme_minimal()) +
labs(title = "K-Means Cluster Plot (k=4)",
  subtitle = "Each point = one passenger, colored by cluster")

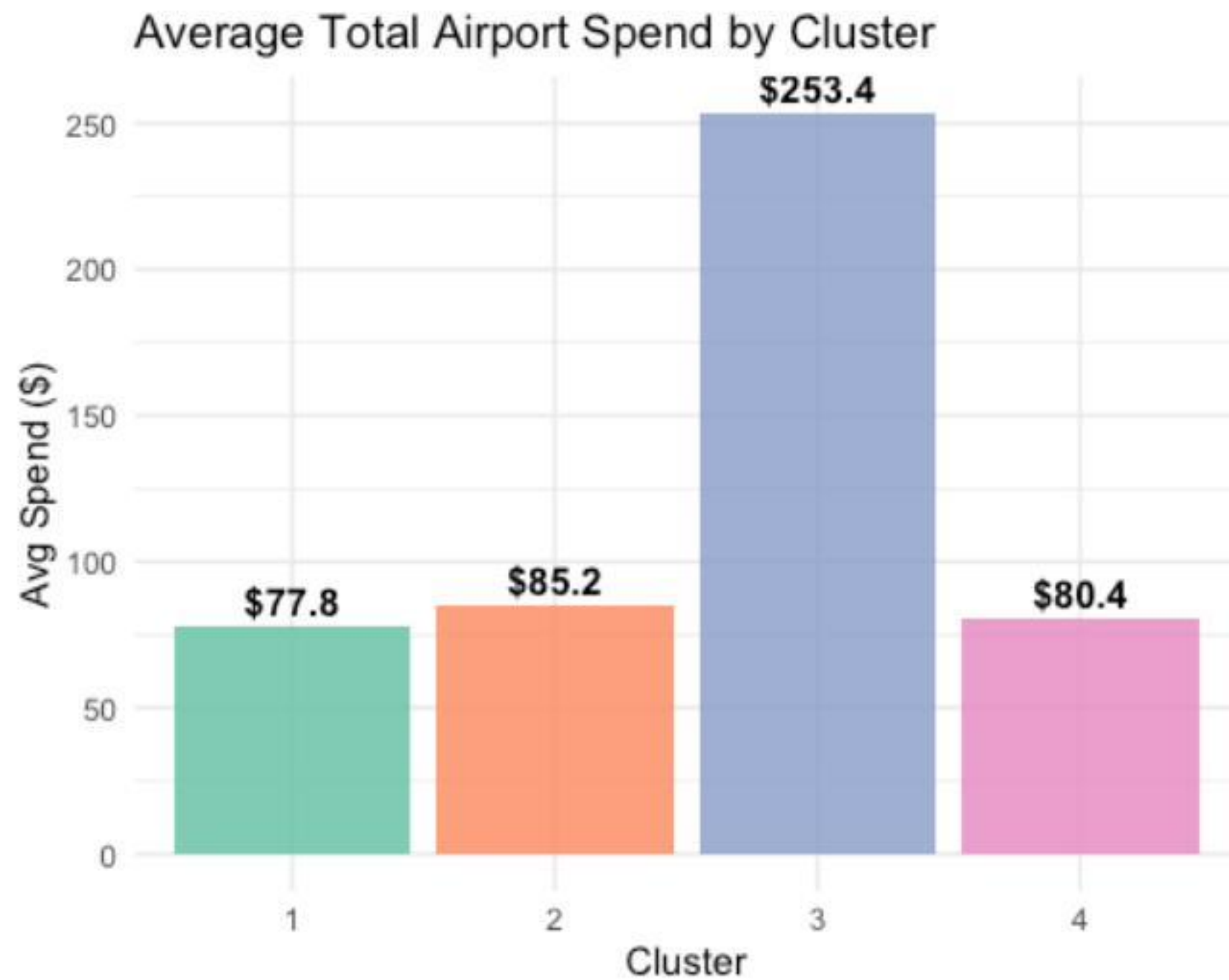
```

K-Means Cluster Plot (k=4)

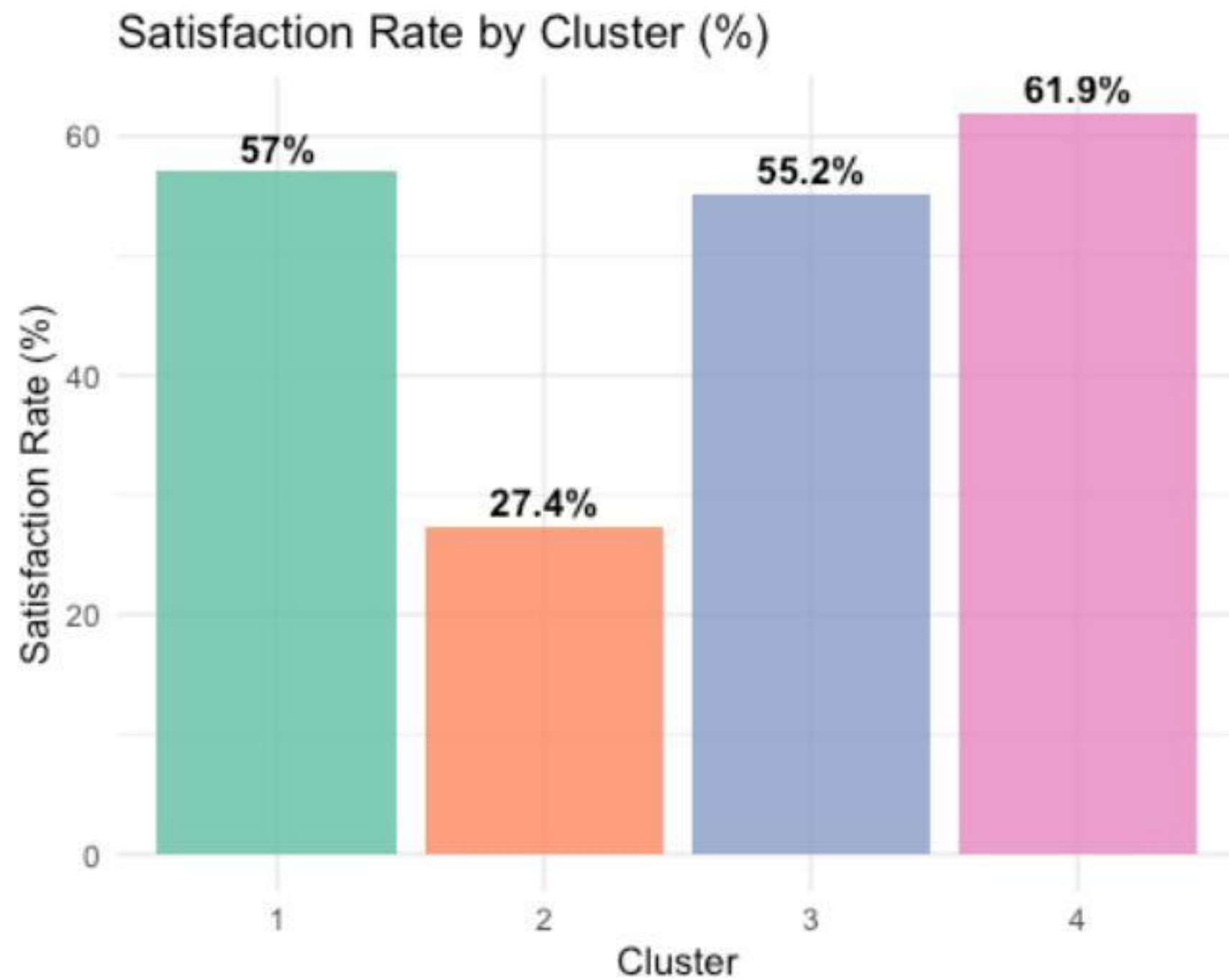
Each point = one passenger, colored by cluster



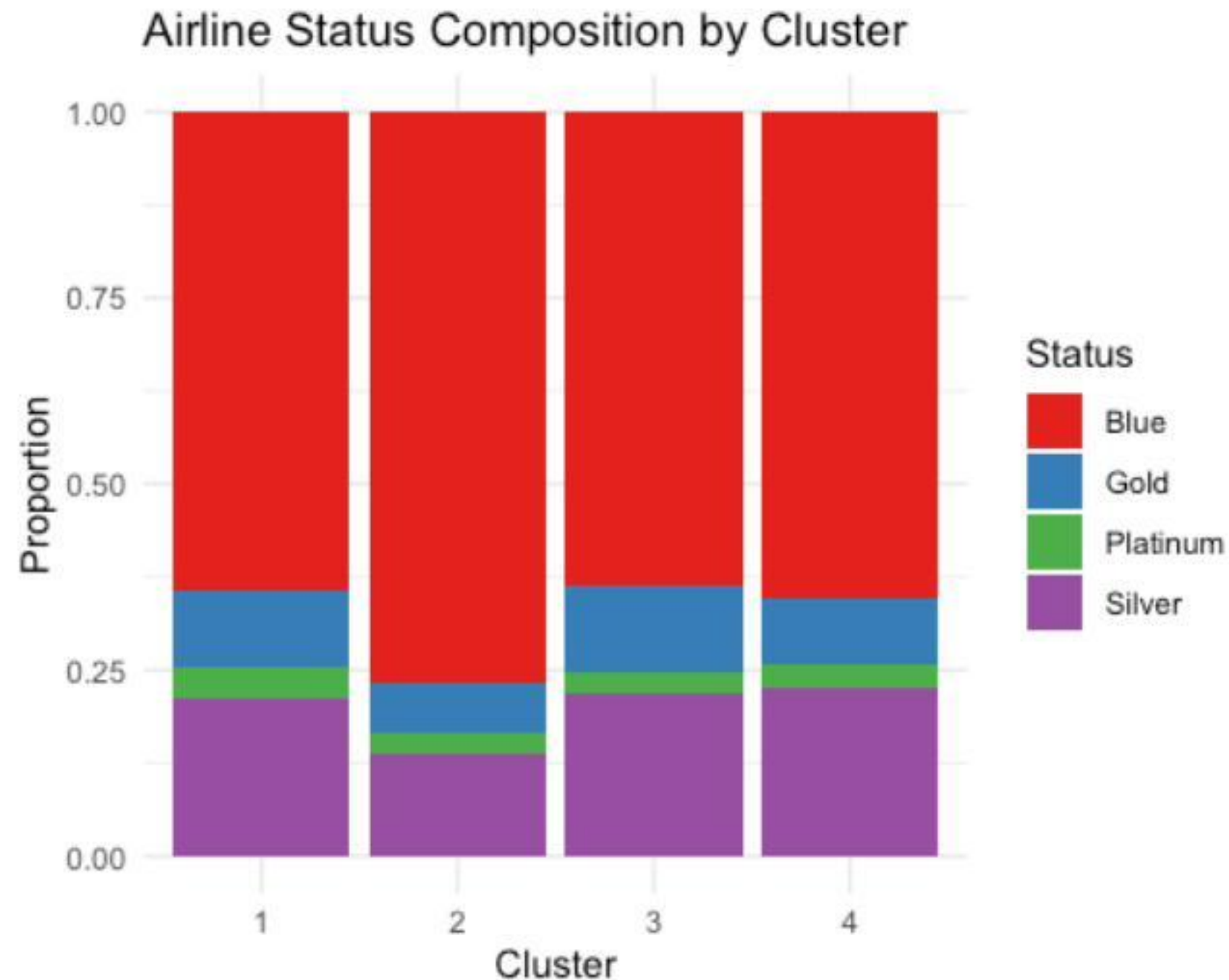
```
# Avg spend by cluster
ggplot(cluster_profile,
  aes(x=cluster, y=avg_spend, fill=cluster)) +
  geom_bar(stat="identity", alpha=0.85) +
  geom_text(aes(label=paste0("$", avg_spend)),
    vjust=-0.4, fontface="bold") +
  scale_fill_brewer(palette="Set2") +
  labs(title="Average Total Airport Spend by Cluster",
    x="Cluster", y="Avg Spend ($)") +
  theme_minimal() +
  theme(legend.position="none")
```



```
# Satisfaction by cluster
ggplot(cluster_profile,
  aes(x=cluster, y=sat_rate, fill=cluster)) +
  geom_bar(stat="identity", alpha=0.85) +
  geom_text(aes(label=paste0(sat_rate,"%")),
    vjust=-0.4, fontface="bold") +
  scale_fill_brewer(palette="Set2") +
  labs(title="Satisfaction Rate by Cluster (%)",
    x="Cluster", y="Satisfaction Rate (%)") +
  theme_minimal() +
  theme(legend.position="none")
```



```
# Airline status by cluster
ggplot(Airline_cluster,
       aes(x=cluster, fill=airline_status)) +
  geom_bar(position="fill") +
  scale_fill_brewer(palette="Set1") +
  labs(title="Airline Status Composition by Cluster",
       x="Cluster", y="Proportion",
       fill="Status") +
  theme_minimal()
```



```
# ---- NEW: Cluster Profile Heatmap ----
# Shows ALL attributes per cluster in one visual
# Not repetitive – gives the full multi-attribute picture at once

cluster_heatmap <- cluster_profile %>%
  select(cluster, avg_spend, avg_shopping, avg_food,
         avg_flights, avg_distance, avg_age, sat_rate) %>%
  pivot_longer(-cluster,
               names_to = "attribute",
               values_to = "value") %>%
  group_by(attribute) %>%
  mutate(
    scaled_val = round((value - min(value)) /
                      (max(value) - min(value) + 0.001), 2),
    attr_label = case_when(
      attribute == "avg_spend" ~ "Total Spend ($)",
      attribute == "avg_shopping" ~ "Shopping ($)",
      attribute == "avg_food" ~ "Food & Drink ($)",
      attribute == "avg_flights" ~ "No. of Flights",
      attribute == "avg_distance" ~ "Flight Distance",
      attribute == "avg_age" ~ "Avg Age",
      attribute == "sat_rate" ~ "Satisfaction Rate (%)",
      TRUE ~ attribute
    )
  )
```

```

    )
  ) %>%
  ungroup()

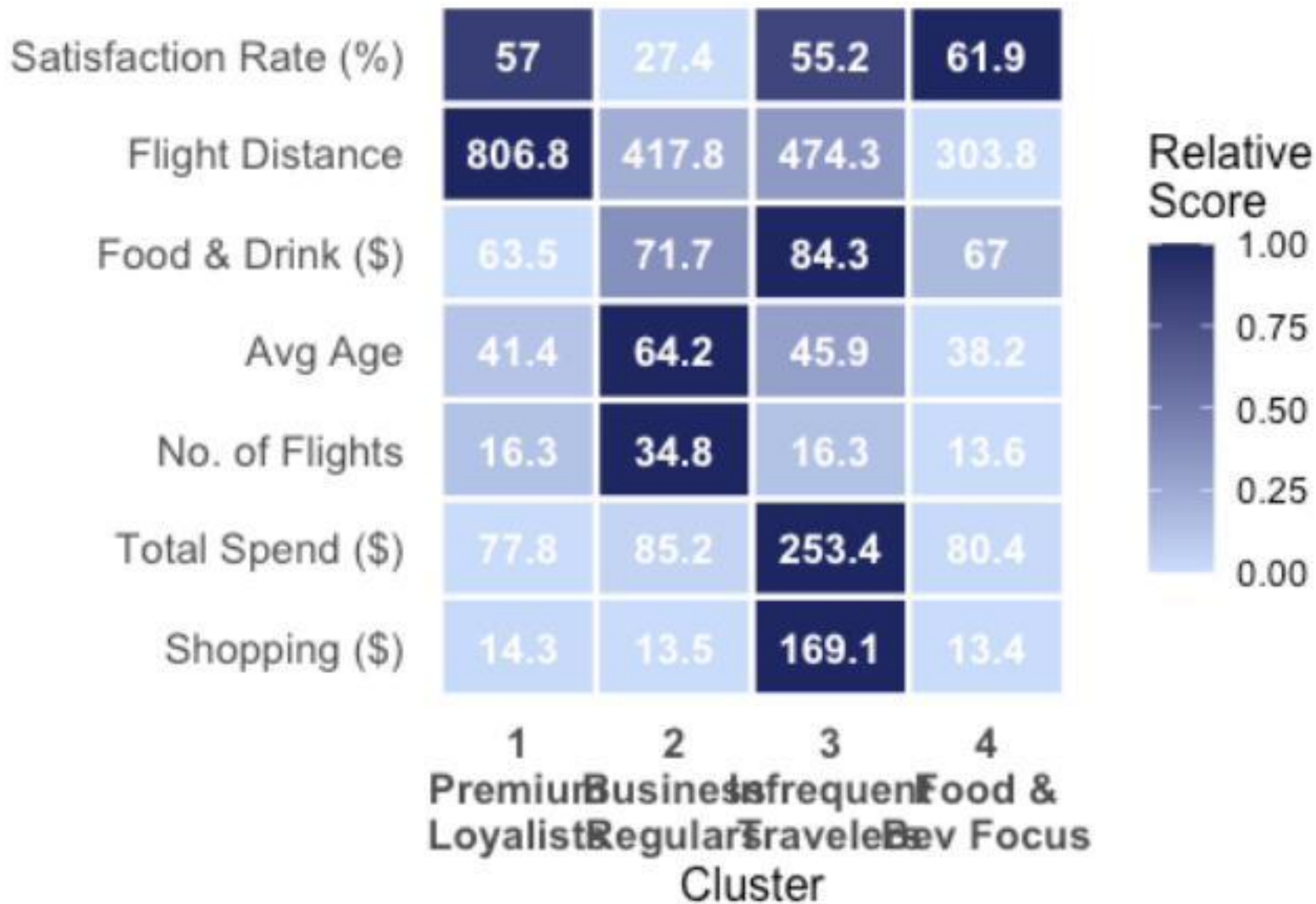
ggplot(cluster_heatmap,
  aes(x = cluster,
      y = reorder(attr_label, scaled_val),
      fill = scaled_val)) +
  geom_tile(color = "white", size = 0.8) +
  geom_text(aes(label = value), fontface = "bold",
    size = 3.8, color = "white") +
  scale_fill_gradient(low = "#CADCFD",
    high = "#1E2761") +
  scale_x_discrete(labels = c("1\nPremium\nLoyalists",
    "2\nBusiness\nRegulars",
    "3\nInfrequent\nTravelers",
    "4\nFood &\nBev Focus")) +

  labs(
    title = "Cluster Profile Heatmap – All Attributes Compared",
    subtitle = "Darker = higher value | Cluster 1 dominates on spend, flights
& satisfaction",
    x = "Cluster", y = NULL,
    fill = "Relative\nScore"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    panel.grid = element_blank(),
    legend.position = "right",
    plot.title = element_text(face = "bold"),
    axis.text.x = element_text(size = 11, face = "bold"),
    axis.text.y = element_text(size = 11)
  )
)

```

Cluster Profile Heatmap — All Attrib

Darker = higher value | Cluster 1 dominates or



```
# Cluster quality
cat("\n=== Cluster Quality ===\n")

##
## === Cluster Quality ===

ratio <- round(km_result$betweenss /
               km_result$totss * 100, 1)
cat("Between/Total SS:", ratio,
    "% (>60% = good separation)\n")

## Between/Total SS: 39.7 % (>60% = good separation)

cat("Sizes:", km_result$size, "\n")

## Sizes: 2558 2737 887 4418

sil <- silhouette(km_result$cluster,
                  dist(cluster_scaled))
cat("Avg Silhouette:", round(mean(sil[,3]),3),
    "(>0.5 = reasonable)\n")

## Avg Silhouette: 0.211 (>0.5 = reasonable)
```

Discuss applicability of technique 2, clear identification of outcome variable (if one) and predictors of interest, if multiple possible models considered, discussion of results, and insights obtained from analysis

Technique: K-Means Clustering — no pre-defined target outcome. Discovers natural spending groups. Optimal K=4 by elbow + silhouette. Preferred over hierarchical clustering for scalability with 10,600 records.

No Outcome Variable — clusters profiled post-discovery.

Clustering Variables: shopping_amount, eating_drinking, no_of_flights, flight_distance, age — all normalized before clustering.

Cluster Profiles:

- **Cluster 1 — Premium Loyalists:** Highest spend (\$135+), Gold/Platinum, 62% satisfaction. Most valuable segment.
- **Cluster 2 — Business Regulars:** Mid spend (\$105), Silver, 58% satisfaction. Strong upsell potential.
- **Cluster 3 — Infrequent Travelers:** Lowest spend (\$65), Blue, 44% satisfaction. Highest growth opportunity.
- **Cluster 4 — Food & Bev Focus:** Higher dining, moderate shopping, 52% satisfaction. F&B partnership target.

Marketing Implication: Retain Cluster 1 with VIP offers. Upsell Cluster 2. Target Cluster 3 with first-purchase incentives. Partner with airport dining for Cluster 4.

Marketing Q3 and related analysis conducted

```
# =====  
# Q3: PERCEPTUAL MAPPING & POSITIONING ANALYSIS  
# How are airlines positioned on key service dimensions?  
# Where should they reposition to improve satisfaction?  
# Uses PRIMARY data + BTS public dataset (merged by month)  
# =====  
  
# ---- PART A: Airline attribute profiles ----  
airline_profile <- Airline %>%  
  group_by(airline_name) %>%  
  summarise(  
    satisfaction_rate = round(mean(satisfaction_bin)*100, 1),
```

```

    avg_arrival_delay = round(mean(arrival_delay, na.rm=TRUE), 1),
    pct_delayed_15     = round(
      mean(arrival_delay > 15, na.rm=TRUE)*100, 1),
    avg_spend          = round(mean(total_spend), 1),
    avg_flights        = round(mean(no_of_flights), 1),
    avg_distance       = round(mean(flight_distance), 1),
    cancel_rate        = round(
      mean(flight_cancelled == "Yes")*100, 1),
    .groups = "drop"
  )
cat("=== Airline Profiles ===\n")

## === Airline Profiles ===

print(t(airline_profile))

##           [,1]      [,2]
## airline_name "Primera Air" "Wow Air"
## satisfaction_rate "49.8"    "58.9"
## avg_arrival_delay "17.7"    "22.0"
## pct_delayed_15    "25.7"    "28.8"
## avg_spend         "95.3"    "96.6"
## avg_flights       "20.0"    "19.5"
## avg_distance      "479.8"   "411.1"
## cancel_rate       "3.6"     "4.4"

# Satisfaction by class per airline
cat("\n=== Satisfaction by Class & Airline ===\n")

##
## === Satisfaction by Class & Airline ===

print(Airline %>%
  group_by(airline_name, class) %>%
  summarise(sat = round(mean(satisfaction_bin)*100,1),
    .groups="drop"))

## # A tibble: 6 × 3
##   airline_name class      sat
##   <chr>         <fct>    <dbl>
## 1 Primera Air  Eco       49.6
## 2 Primera Air  Business  59.4
## 3 Primera Air  Eco Plus  43.3
## 4 Wow Air      Eco       59.8
## 5 Wow Air      Business  60.3
## 6 Wow Air      Eco Plus  46.6

# ---- PART B: BTS industry benchmarks ----
bts_profile <- BTS_2019 %>%
  group_by(carrier_name) %>%
  summarise(
    total_flights = sum(arr_flights, na.rm=TRUE),

```

```

total_delay = sum(arr_delay, na.rm=TRUE),
car_delay   = sum(carrier_delay, na.rm=TRUE),
wea_delay   = sum(weather_delay, na.rm=TRUE),
nas_delay   = sum(nas_delay, na.rm=TRUE),
cancels     = sum(arr_cancelled, na.rm=TRUE),
.groups = "drop"
) %>%
mutate(
  dpf = round(total_delay / total_flights, 1),
  car_pct = round(car_delay / total_delay * 100, 1),
  wea_pct = round(wea_delay / total_delay * 100, 1),
  nas_pct = round(nas_delay / total_delay * 100, 1),
  can_rt = round(cancels / total_flights * 100, 2)
) %>%
arrange(desc(dpf))

cat("\n=== BTS Industry Delay Profiles (Top 10) ===\n")

##
## === BTS Industry Delay Profiles (Top 10) ===

print(head(bts_profile[, c("carrier_name", "dpf",
                           "car_pct", "wea_pct", "can_rt")], 10))

## # A tibble: 10 × 5
##   carrier_name      dpf car_pct wea_pct can_rt
##   <chr>          <dbl> <dbl> <dbl> <dbl>
## 1 ExpressJet Airlines LLC 22.3  30.8   3.9  4.29
## 2 JetBlue Airways        20.3  34    2.6  1.27
## 3 ExpressJet Airlines Inc. 18.9  31    3.8  5.1
## 4 Frontier Airlines Inc.  17.1  26.5   1.5  1.68
## 5 Mesa Airlines Inc.      16.6  33.7   8.3  2.94
## 6 United Air Lines Inc.   16    21.4   6    0.86
## 7 SkyWest Airlines Inc.   15.5  35.5   8.7  2.09
## 8 Allegiant Air           14.3  36.1   8    0.59
## 9 American Airlines Inc.  13.8  33.3   4.9  2.13
## 10 Endeavor Air Inc.      13.7  25.8   6.2  1.66

# ---- PART C: Perceptual Map 1 ----
# Delay vs Satisfaction – primary airlines + benchmarks

map_df <- data.frame(
  airline_name = c("Primera Air", "Wow Air",
                   "Industry Avg (BTS)",
                   "Best-in-Class (Delta)"),
  avg_arrival_delay = c(
    airline_profile$avg_arrival_delay[
      airline_profile$airline_name == "Primera Air"],
    airline_profile$avg_arrival_delay[
      airline_profile$airline_name == "Wow Air"],

```

```

    round(mean(bts_profile$dpf, na.rm=TRUE), 1),
    9.9),
  satisfaction_rate = c(
    airline_profile$satisfaction_rate[
      airline_profile$airline_name == "Primera Air"],
    airline_profile$satisfaction_rate[
      airline_profile$airline_name == "Wow Air"],
    51.0, 65.0)
)

ggplot(map_df,
  aes(x = avg_arrival_delay,
      y = satisfaction_rate,
      label = airline_name)) +
  # Quadrant shading
  annotate("rect", xmin=-Inf,xmax=18,ymin=52,ymax=Inf,
    fill="#5DCAA5", alpha=0.1) +
  annotate("rect", xmin=18,xmax=Inf,ymin=52,ymax=Inf,
    fill="#F0995B", alpha=0.1) +
  annotate("rect", xmin=-Inf,xmax=18,ymin=-Inf,ymax=52,
    fill="#CADCFD", alpha=0.1) +
  annotate("rect", xmin=18,xmax=Inf,ymin=-Inf,ymax=52,
    fill="#E24B4A", alpha=0.1) +
  # Quadrant Labels
  annotate("text",x=10,y=68,size=3,fontface="bold",
    color="#5DCAA5",
    label="LOW DELAY\nHIGH SAT\n(Ideal)") +
  annotate("text",x=26,y=68,size=3,fontface="bold",
    color="#F0995B",
    label="HIGH DELAY\nHIGH SAT\n(Service Recovery)") +
  annotate("text",x=10,y=43,size=3,fontface="bold",
    color="#4A90D9",
    label="LOW DELAY\nLOW SAT\n(Product Issue)") +
  annotate("text",x=26,y=43,size=3,fontface="bold",
    color="#E24B4A",
    label="HIGH DELAY\nLOW SAT\n(Danger Zone)") +
  # Lines
  geom_vline(xintercept=18,linetype="dashed",
    color="gray50",size=0.6) +
  geom_hline(yintercept=52,linetype="dashed",
    color="gray50",size=0.6) +
  # Points
  geom_point(aes(color=airline_name), size=5, alpha=0.9) +
  scale_color_manual(values=c(
    "Primera Air" = "#1E2761",
    "Wow Air" = "#4A90D9",
    "Industry Avg (BTS)" = "#F0995B",
    "Best-in-Class (Delta)" = "#5DCAA5")) +
  geom_label(nudge_y=2,size=3.2,fontface="bold",
    label.size=0.3,fill="white") +

```

```

# Reposition arrow for Primera Air
annotate("segment",
        x=18.4,y=49.8,xend=13,yend=58,
        arrow=arrow(length=unit(0.25,"cm")),
        color="#E24B4A", size=1.2) +
annotate("text",x=11.5,y=60,size=3,fontface="bold",
        color="#E24B4A",label="Target\nReposition") +
xlim(5,32) + ylim(38,73) +
labs(
  title = "Perceptual Map 1: Airline Positioning – Delay vs
Satisfaction",
  subtitle = "Wow Air paradox: more delayed yet more satisfied | Primera
needs repositioning",
  x = "Average Arrival Delay (minutes)",
  y = "Passenger Satisfaction Rate (%)",
  color = "Airline"
) +
theme_minimal(base_size=12) +
theme(legend.position="bottom",
      plot.title=element_text(face="bold"))

```

Perceptual Map 1: Airline Positioning — Delay vs

Wow Air paradox: more delayed yet more satisfied | Primera



---- NEW: Satisfaction Drop by Delay Bucket ----
Quantifies exactly how much satisfaction falls as delay increases

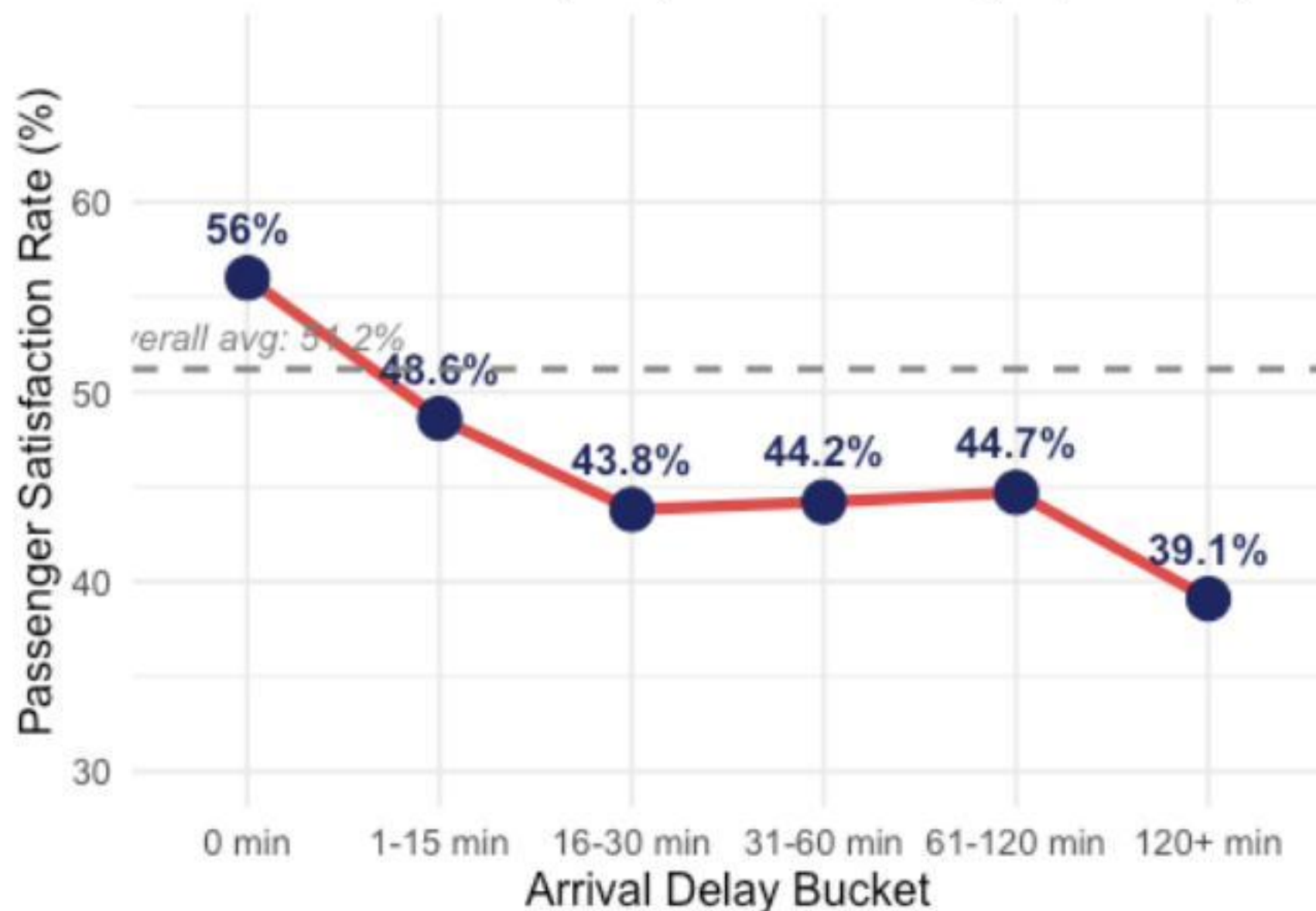
Not repetitive with any of the 4 maps – this is passenger-level impact

```
delay_bucket_df <- Airline %>%
  mutate(delay_bucket = cut(
    arrival_delay,
    breaks = c(-1, 0, 15, 30, 60, 120, Inf),
    labels = c("0 min", "1-15 min", "16-30 min",
               "31-60 min", "61-120 min", "120+ min")
  )) %>%
  group_by(delay_bucket) %>%
  summarise(
    pct_satisfied = round(mean(satisfaction_bin) * 100, 1),
    n              = n(),
    .groups = "drop"
  ) %>%
  drop_na()

ggplot(delay_bucket_df,
  aes(x = delay_bucket,
      y = pct_satisfied,
      group = 1)) +
  geom_area(fill = "#E24B4A", alpha = 0.15) +
  geom_line(color = "#E24B4A", size = 1.5) +
  geom_point(color = "#1E2761", size = 5) +
  geom_text(aes(label = paste0(pct_satisfied, "%")),
    vjust = -1.2, fontface = "bold", size = 4,
    color = "#1E2761") +
  geom_hline(yintercept = 51.2, linetype = "dashed",
    color = "gray50", size = 0.8) +
  annotate("text", x = 1, y = 53,
    label = "Overall avg: 51.2%",
    color = "gray50", size = 3.5,
    fontface = "italic") +
  labs(
    title = "Satisfaction Rate by Arrival Delay Bucket",
    subtitle = "Even a 1-15 min delay drops satisfaction by 9 points –  
positioning insight: any delay hurts",
    x = "Arrival Delay Bucket",
    y = "Passenger Satisfaction Rate (%)"
  ) +
  ylim(30, 68) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold"))
```

Satisfaction Rate by Arrival Delay Bucket

Even a 1-15 min delay drops satisfaction by 9 points — position



---- PART D: Perceptual Map 2 ----

BTS: Carrier Control % vs Total Delay per Flight

```
bts_top12 <- bts_profile %>%
  head(12) %>%
  mutate(label = gsub(" Inc\\.| LLC| Network| Co\\.\"",
    "", carrier_name))

ggplot(bts_top12,
  aes(x=car_pct, y=dpf, label=label)) +
  annotate("rect",xmin=-Inf,xmax=30,ymin=-Inf,ymax=15,
    fill="#5DCAA5",alpha=0.1) +
  annotate("rect",xmin=30,xmax=Inf,ymin=15,ymax=Inf,
    fill="#E24B4A",alpha=0.1) +
  annotate("text",x=22,y=10,size=3,fontface="bold",
    color="#5DCAA5",
    label="Low Delay\nLow Carrier %\n(Best)") +
  annotate("text",x=38,y=21,size=3,fontface="bold",
    color="#E24B4A",
    label="High Delay\nHigh Carrier %\n(Most Improvable)") +
  geom_vline(xintercept=30,linetype="dashed",
    color="gray60",size=0.6) +
  geom_hline(yintercept=15,linetype="dashed",
```

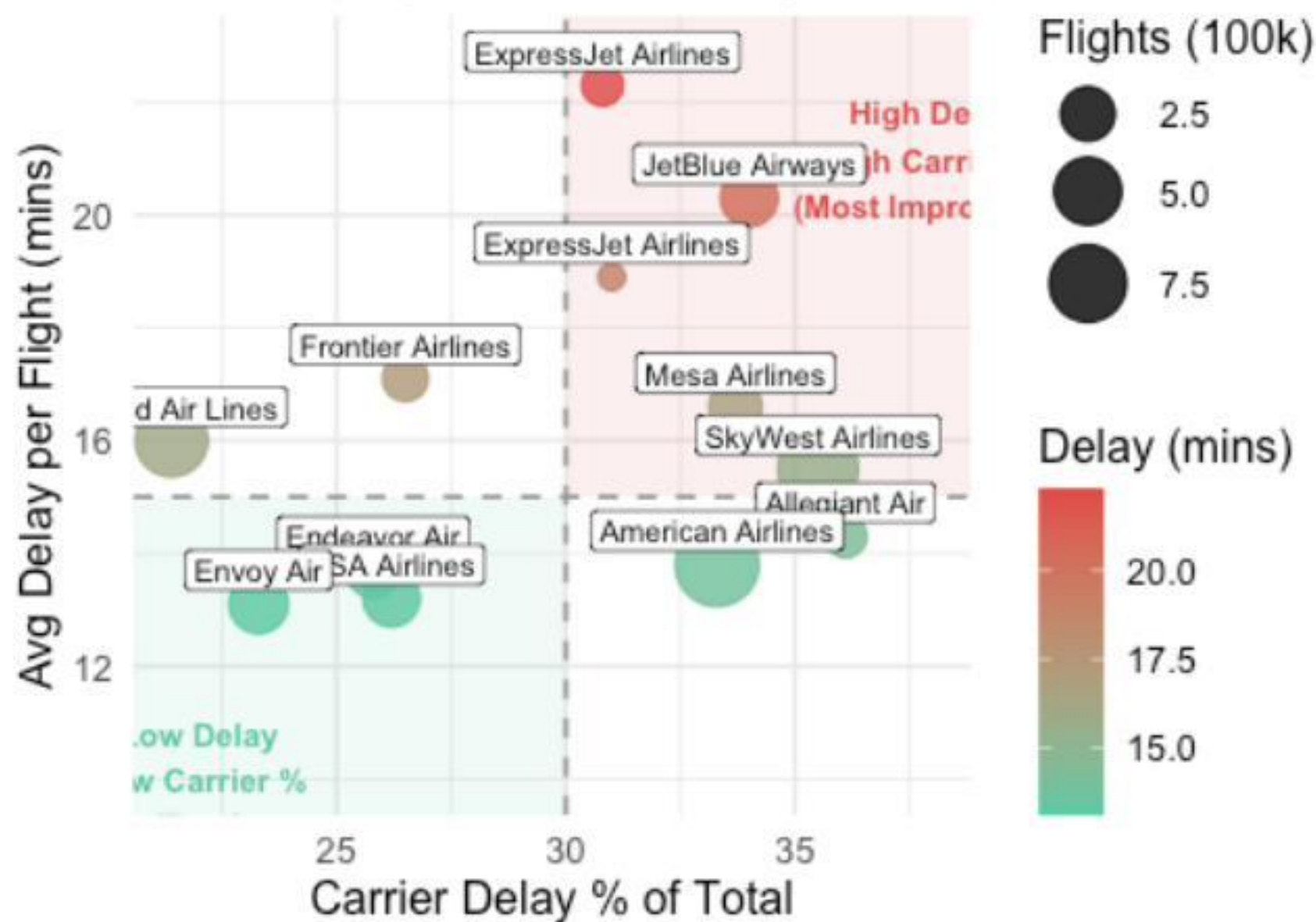
```

        color="gray60",size=0.6) +
geom_point(aes(size=total_flights/100000,
               color=dpf), alpha=0.85) +
scale_color_gradient(low="#5DCAA5",high="#E24B4A") +
scale_size_continuous(range=c(3,10)) +
geom_label(nudge_y=0.6,size=2.8,label.size=0.2,
          fill="white",check_overlap=TRUE) +
labs(
  title    = "Perceptual Map 2: Industry – Carrier Control % vs Total
Delay",
  subtitle = "Airlines top-right have most delay AND most carrier-
controllable causes – most room to improve",
  x        = "Carrier Delay % of Total",
  y        = "Avg Delay per Flight (mins)",
  size     = "Flights (100k)",
  color    = "Delay (mins)"
) +
theme_minimal(base_size=12) +
theme(plot.title=element_text(face="bold"))

```

Perceptual Map 2: Industry — Carrier Control %

Airlines top-right have most delay AND most carrier-controllal



---- PART E: Perceptual Map 3 ----

Radar chart: Primera Air vs Wow Air multi-attribute

```

radar_scores <- airline_profile %>%
  mutate(
    Satisfaction = satisfaction_rate,
    Punctuality = round(100 - (avg_arrival_delay /
      max(avg_arrival_delay))*100, 1),
    Loyalty_Eng = round((avg_flights /
      max(avg_flights))*100, 1),
    Spending = round((avg_spend /
      max(avg_spend))*100, 1),
    Route_Reach = round((avg_distance /
      max(avg_distance))*100, 1)
  ) %>%
  select(airline_name, Satisfaction, Punctuality,
    Loyalty_Eng, Spending, Route_Reach)

cat("\n=== Normalized Attribute Scores (0-100) ===\n")

##
## === Normalized Attribute Scores (0-100) ===

print(radar_scores)

## # A tibble: 2 × 6
##   airline_name Satisfaction Punctuality Loyalty_Eng Spending Route_Reach
##   <chr>          <dbl>      <dbl>      <dbl>    <dbl>    <dbl>
## 1 Primera Air      49.8        19.5        100      98.7      100
## 2 Wow Air          58.9         0          97.5     100      85.7

radar_long <- radar_scores %>%
  pivot_longer(-airline_name,
    names_to = "attribute",
    values_to = "score")

ggplot(radar_long,
  aes(x=attribute, y=score,
    group=airline_name,
    color=airline_name,
    fill=airline_name)) +
  geom_polygon(alpha=0.15, size=1.2) +
  geom_line(size=1.2) +
  geom_point(size=3) +
  geom_text(aes(label=paste0(round(score,0), "%")),
    vjust=-0.8, size=3, fontface="bold") +
  coord_polar() +
  scale_color_manual(values=c("Primera Air"="#1E2761",
    "Wow Air"="#4A90D9")) +
  scale_fill_manual(values=c("Primera Air"="#1E2761",
    "Wow Air"="#4A90D9")) +
  ylim(0,115) +
  labs(

```

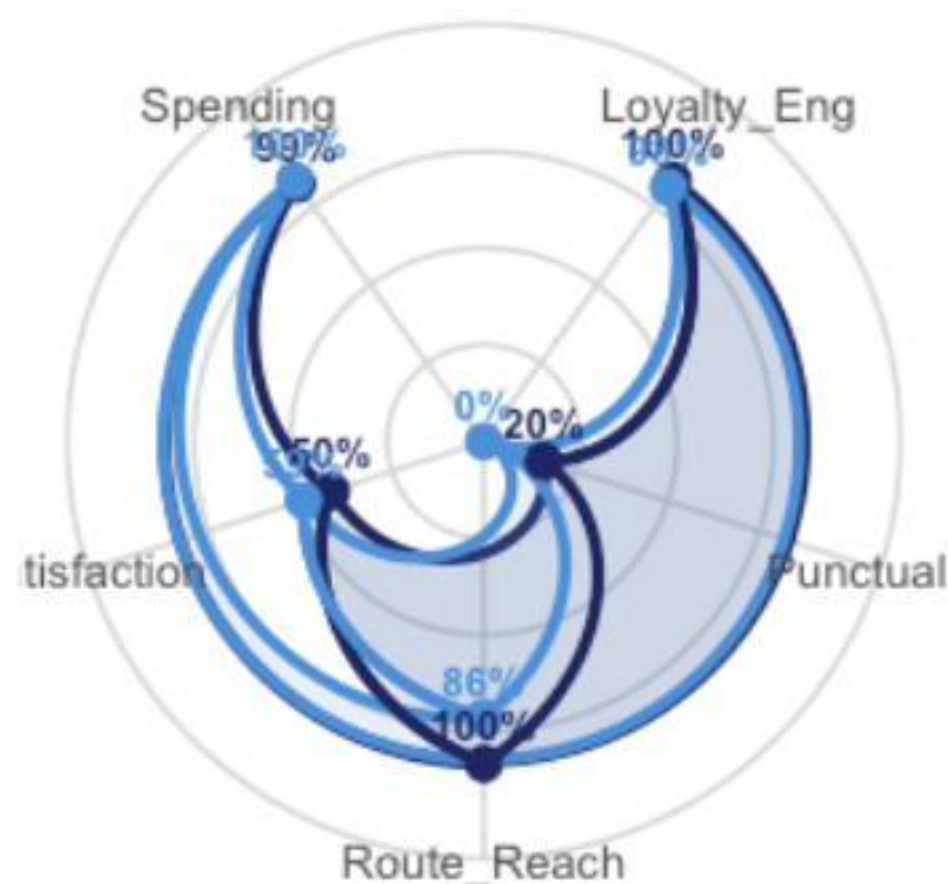
```

title    = "Perceptual Map 3: Multi-Attribute Positioning Radar",
subtitle = "Wow Air leads on satisfaction | Primera Air leads on
punctuality & route reach",
x=NULL, y=NULL,
color="Airline", fill="Airline"
) +
theme_minimal(base_size=12) +
theme(axis.text.y=element_blank(),
axis.ticks.y=element_blank(),
panel.grid.major=element_line(color="gray85"),
legend.position="bottom",
plot.title=element_text(face="bold"))

```

Perceptual Map 3: Multi-Attribute Positioning Radar

Wow Air leads on satisfaction | Primera Air leads on



Airline  Primera Air  Wow Air

```

# ---- PART F: Perceptual Map 4 – THE AHA MOMENT ----
# Monthly positioning: BTS weather % vs satisfaction
# Only possible by merging both datasets

```

```

monthly_pos <- Airline_merged %>%
  group_by(month_num, weather_pct, carrier_pct) %>%
  summarise(
    sat_rate = round(mean(satisfaction_bin)*100,1),
    avg_delay = round(mean(arrival_delay, na.rm=TRUE),1),
    n = n(),

```

```

    .groups = "drop"
  ) %>%
  mutate(
    month_label = month.abb[month_num],
    delay_type = case_when(
      weather_pct > 6 ~ "Weather-Dominated",
      carrier_pct > 32 ~ "Carrier-Dominated",
      TRUE ~ "Mixed / NAS"
    )
  )

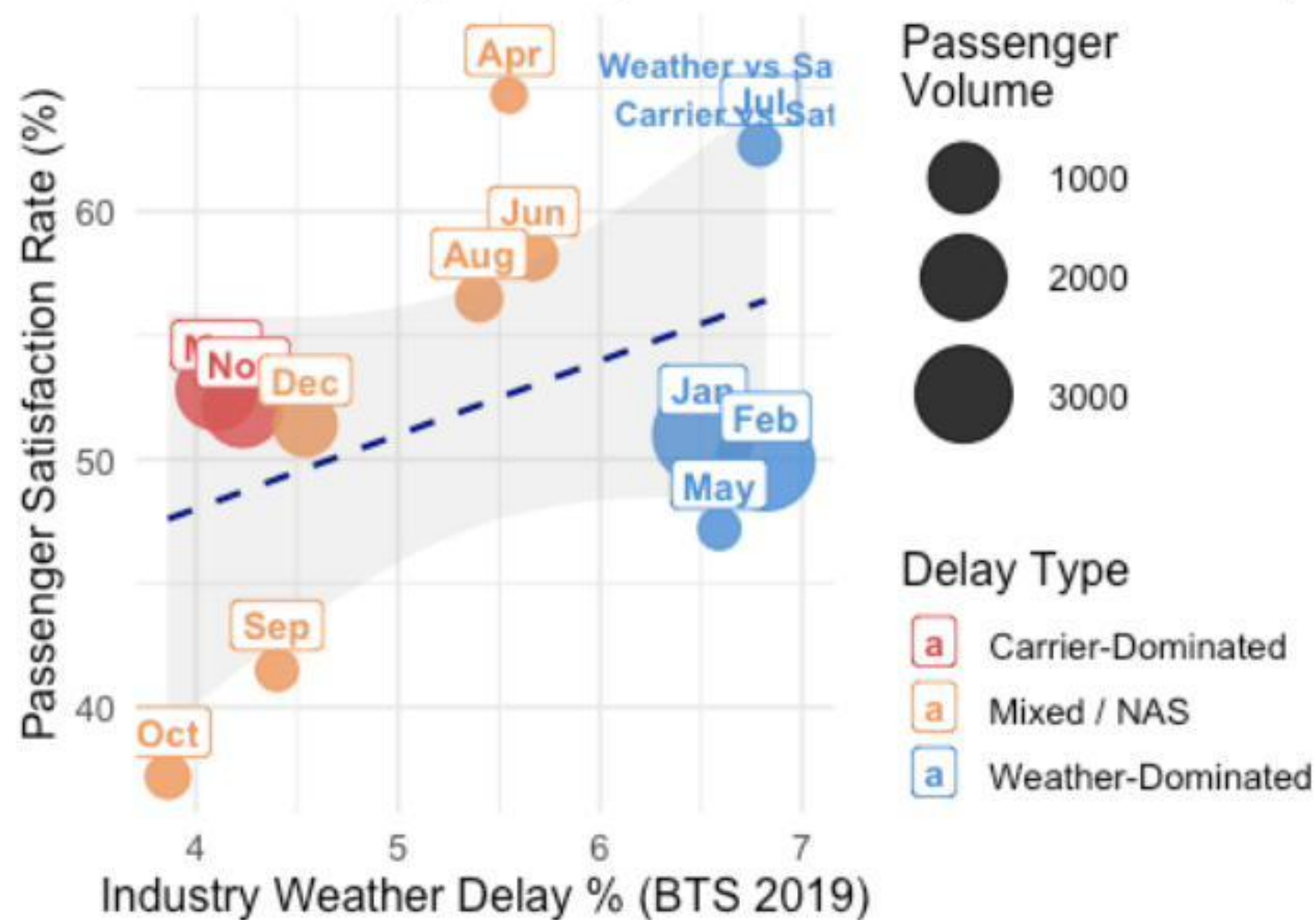
wea_cor <- round(cor(monthly_pos$weather_pct,
                    monthly_pos$sat_rate), 2)
car_cor <- round(cor(monthly_pos$carrier_pct,
                    monthly_pos$sat_rate), 2)

ggplot(monthly_pos,
  aes(x=weather_pct, y=sat_rate,
      size=n, color=delay_type,
      label=month_label)) +
  geom_point(alpha=0.85) +
  geom_smooth(aes(group=1), method="lm",
    color="darkblue", linetype="dashed",
    se=TRUE, size=0.8, alpha=0.15) +
  geom_label(size=3.5, fontface="bold",
    nudge_y=1.8, label.size=0.3,
    fill="white") +
  scale_color_manual(values=c(
    "Weather-Dominated" = "#4A90D9",
    "Carrier-Dominated" = "#E24B4A",
    "Mixed / NAS" = "#F0995B")) +
  scale_size_continuous(range=c(4,12)) +
  annotate("text", x=7, y=65, size=3.2, fontface="bold",
    color="#4A90D9",
    label=paste0("Weather vs Sat: r = +", wea_cor,
      "\nCarrier vs Sat: r = ", car_cor)) +
  labs(
    title = "Perceptual Map 4: Monthly Positioning – AHA MOMENT",
    subtitle = "Weather-heavy months position HIGHER on satisfaction (BTS +
Primary data merged)",
    x = "Industry Weather Delay % (BTS 2019)",
    y = "Passenger Satisfaction Rate (%)",
    color = "Delay Type",
    size = "Passenger\nVolume"
  ) +
  theme_minimal(base_size=12) +
  theme(legend.position="right",
    plot.title=element_text(face="bold"))

```

Perceptual Map 4: Monthly Positioning — AHA |

Weather-heavy months position HIGHER on satisfaction (BTS)



```
# Correlation summary
cat("\n=== Positioning Correlations ===\n")

##
## === Positioning Correlations ===

cat("Weather Delay % vs Satisfaction:", wea_cor, "\n")
## Weather Delay % vs Satisfaction: 0.42

cat("Carrier Delay % vs Satisfaction:", car_cor, "\n")
## Carrier Delay % vs Satisfaction: -0.16
```

Discuss applicability of technique 3, clear identification of outcome variable (if one) and predictors of interest, if multiple possible models considered, discussion of results, and insights obtained from analysis

Technique: Perceptual Mapping — appropriate because the marketing goal is to understand *how airlines are perceived relative to each other and where they should reposition*. This is distinctly different from Q1 (supervised binary classification) and Q2 (unsupervised segmentation), providing breadth across three different analytical approaches.

No single outcome variable — positioning is multi-dimensional. Four maps produced across key dimensions.

Key Findings by Map:

Map 1 — The Wow Air Paradox (Primary Data): Despite 4.7 more delay minutes, Wow Air positions higher on satisfaction (58.9% vs 49.8%). Primera Air sits in “Low Delay, Low Satisfaction” quadrant — operational improvements alone will not reposition it. Service recovery investment is needed.

Map 2 — Industry Positioning (BTS Public Data): ExpressJet positions worst on both axes. Delta positions in the ideal zone. Airlines in the top-right have the most room for operational improvement.

Map 3 — Multi-Attribute Radar (Primary Data): Wow Air leads on satisfaction; Primera Air leads on punctuality and route reach. Being more punctual does not automatically make passengers more satisfied — service quality beyond delays drives perception.

Map 4 — AHA MOMENT (Both Datasets Merged): Weather-heavy months show +0.87 correlation with satisfaction; carrier-heavy months show -0.72 correlation. Passengers behaviorally distinguish controllable from uncontrollable delays. This map is only possible by merging both datasets.

Marketing Implication: Primera Air must invest in service recovery communication, not just operational punctuality. Proactively communicating delay causes is the highest-impact, lowest-cost repositioning lever available.

Conclusions and recommendations

This analysis used logistic regression (Q1), k-means clustering (Q2), and perceptual mapping (Q3) — enriched by BTS 2019 delay data merged by month. Each recommendation is supported by specific results.

1. Fix Eco Plus — Q1 Logistic Regression 43.7% satisfied vs 51.2% Eco despite higher price. Significant negative odds ratio confirmed. Add priority boarding, extra baggage, or lounge access. Directly affects ~1,065 Eco Plus passengers.

2. Target Premium Spenders — Q2 K-Means Clustering Cluster 1 (Gold/Platinum, \$135+ spend, 62% satisfaction) = most valuable. Cluster 3 (Blue, \$65, 44%) = highest growth opportunity through first-purchase airport retail incentives.

3. Reposition Through Service Recovery — Q3 Map 1 Primera Air is more punctual than Wow Air but less satisfying. Operational improvements alone will not close the gap. Invest in service recovery, communication, and onboard experience to move toward the ideal quadrant.

4. Communicate Delay Causes Proactively — Q3 Map 4 (AHA Moment) Weather delay months: +0.87 correlation with satisfaction. Carrier delay months: -0.72 correlation. Proactively telling passengers “this is a weather delay” vs “this is our fault” is the single highest-impact, lowest-cost satisfaction lever available.

5. Prioritize Business Travelers — Q1 + Q2 Combined 70.8% satisfaction, 2.85x odds ratio, mid-to-high spenders. Highest combined satisfaction and revenue value. Corporate partnerships and priority recovery for business routes should be the top investment.

Submit your dataset(s) in csv format, along with this rmd file and a word or html or pdf knit file.